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(54) **GENOMIC ALTERATION OF PLANT
GERMLINE**

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None
See application file for complete search history.

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(57) **ABSTRACT**

Compositions containing chimeric RNA molecules which
comprise meristem targeting sequences that are fused to
RNA cargo sequences that include gene editing molecules
are provided. Methods of using the compositions to effi-
ciently edit plant genomes without intervening tissue culture
steps are also provided. The solutions described here relate
to engineered RNA molecules useful in producing plants
with altered genomes. As such, it relates to substantially
purified compositions, vectors, systems, as well as genomes
of plants.

23 Claims, 2 Drawing Sheets

Specification includes a Sequence Listing.

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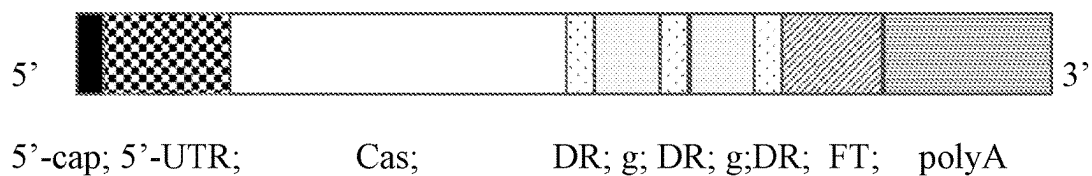
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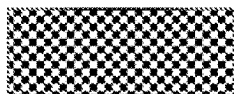


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5'-cap



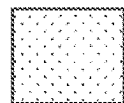
5'-UTR



Cas



DR



g (gRNA)



FT



polyA

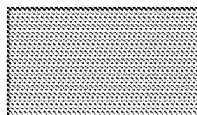
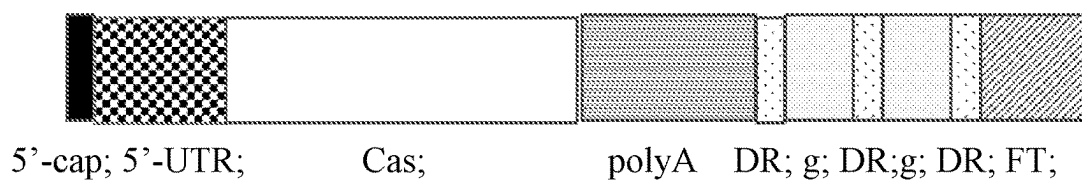


FIGURE 1A

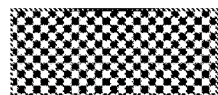


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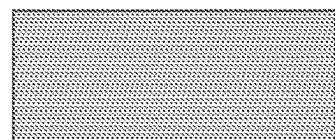
5'-UTR



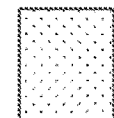
Cas



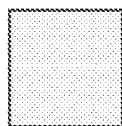
polyA



DR



g (gRNA)



FT

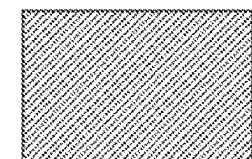


FIGURE 1B

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**GENOMIC ALTERATION OF PLANT
GERMLINE****CROSS-REFERENCE TO RELATED
APPLICATIONS**

This international patent application claims the benefit of U.S. provisional patent application No. 62/924,542, filed Oct. 22, 2019 and incorporated herein by reference in its entirety.

**ELECTRONICALLY REFERENCE TO
SEQUENCE LISTING SUBMITTED
ELECTRONICALLY**

The content of the electronically submitted sequence listing in ASCII text file (Name: 10068_SEQ LST_ST25.txt; Size: 102655 bytes; and Date of Creation: Oct. 22, 2020) filed with the application is incorporated herein by reference in its entirety.

BACKGROUND

Development of new and improved varieties of plants requires a genetically diverse parental pool. Traditional breeding programs are based on genetic variation that originates from exotic germplasm or from random mutagenesis. Selected individuals with potentially advantageous genetic traits are backcrossed into elite germplasm to develop improved varieties.

With a growing understanding of plant genetics, many targets emerge for possible genetic modifications useful in making improved plant varieties. Yet traditional methods of random mutagenesis are time consuming and do not provide a convenient way to explore the full spectrum of potential benefit of genetic variation of candidate loci. Other methods like transgenesis or genome editing are more promising.

A drawback of specific genomic intervention, such as by genome editing, is our limited current ability to directly modify the genome of elite germplasm of the species of interest. Genome editing reagents are most often delivered to transformable rather than elite germplasm, which needs to be followed by prolonged backcrossing into commercial germplasm before the phenotypic impact of individual edits can be assessed. The editing methods often require tissue culture and plant regeneration, which requires specific skills and equipment, and adds significant time and expense to the entire process. The methods are very complicated for most plant species, sometimes requiring use of morphogenic regulators to facilitate successful gene editing reagent delivery using biolistic- or Agrobacterium-mediated methods. This is followed by a long process of selecting the putative edited cells and regenerating the edited plants via a complex tissue culture process that is specific to each genotype for plant species of interest. The dedifferentiation required to produce regenerable callus using tissue culture often triggers seemingly random epigenetic modifications, which further complicates any phenotypic analyses of primary transformants and their progeny.

A need remains for robust and efficient reagents and methods for performing targeted genetic editing in plants. Ideally, the solutions are broadly applicable or easily adaptable to different species and varieties within each species. Bypassing callus induction and/or tissue culture is preferable, to reduce the time and resources required to produce

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edited events and to produce many targeted genetic variants plus their combinations in all relevant elite genetic backgrounds.

SUMMARY

The solutions described here relate to engineered RNA molecules useful in producing plants with altered genomes. As such, it relates to substantially purified compositions, vectors, systems, as well as methods, seeds, pollen, and plants useful at various steps in altering genomes of plants.

In their use, the RNA molecules are often needed in a substantially purified form. The RNAs are generally chimeric, meaning that they are made up of at least two different fused segments. One segment comprises a cargo RNA sequence, and another segment comprises a meristem transport RNA sequence.

The cargo segment is made up of RNA that, once inside meristematic cells, carries out the genome alteration function. In various embodiments, the cargo segment can be made up one or more of different sequences needed for the assembly in the plant cell cytosol of the genome-altering function, i.e. it has one or more DNA-modifying components. The DNA modifying components are typically RNA-guided nuclease components, RNAi, a TALE, zinc finger, or meganuclease sequences. RNA-guided nuclease systems typically require at least one polypeptide nuclease effector and one or more guide RNAs. In some embodiments, the cargo segment has an expressible coding sequence of a polypeptide nuclease effector (e.g. Cas9, Cas12a, or Cas12i), such that the RNA is translated when inside a plant cell. In some embodiments, the cargo segment comprises guide RNAs that are flanked by processing elements designed so that, within a plant cell cytosol, they are excised from the chimeric molecule and function in conjunction with a polypeptide nuclease effector present in the same cell. In some embodiments, the same RNA molecule comprises both the effector polynucleotide-encoding sequence and one or more guide RNAs. In these cases, the guide RNA processing elements can be made up of direct repeat sequences of the bacterial CRISPR array of the RNA-guided polypeptide.

The meristem transport segment is made up of a sequence that allows for transport of a chimeric RNA through the plant (e.g., through the phloem of the vascular system) and into the meristem tissues or meristem cells. The transport segment sequence can occur in any RNA found in the plant vascular system that transits from the tissue/cell of origin to the meristem. In one embodiment, the transport segment sequence is generally based on Flowering Time (FT) genes of plants, and they sometimes correspond to fragments of FT transcripts. Flowering Time (FT) gene products are also referred to as "florigen." The chimeric RNAs are often arranged so that the meristem transport segment is often located 3' of the cargo segment. In another embodiment, the chimeric RNAs are arranged so that the meristem transport segment (MTS) is located 3' of the protein coding segment (e.g., a segment encoding an RNA-guided nuclease) in the chimeric RNA.

The RNAs can be used in methods of producing plants with altered genomes. Accordingly, a subject plant is contacted with RNAs as described, so that the RNAs typically reach the phloem of the plant. This step may be carried out at the vegetative stage of the plant life cycle. Germline cells of the treated plant and their progeny will have the genome alterations intended to be made by the introduced RNA. In certain embodiments, germline cells of the treated plant and their daughter cells will have the intended genome

alterations encoded by the introduced RNA prior to transitioning to reproductive development.

In certain embodiments, a composition comprising a substantially purified RNA molecule made up of a cargo segment fused to a meristem transport segment is provided. In certain embodiments, the cargo segment comprises a DNA-modifying component. In certain embodiments, the DNA-modifying component is selected from an RNA-guided nuclease component, an RNAi, a TALE, a zinc finger, and a meganuclease. In certain embodiments, the RNA-guided nuclease component comprises an RNA-guided polypeptide encoding sequence. In certain embodiments, the RNA-guided polypeptide encoding sequence can be translated if present in a plant cell cytosol. In certain embodiments, the meristem transport segment comprises an FT-derived sequence. In certain embodiments, the FT-derived sequence is a fragment of an FT transcript. In certain embodiments, the meristem transport segment is located 3' of the cargo segment. In any of the aforementioned embodiments, the composition further comprise RNase inhibitors. A method of producing a plant with an altered genome, comprising contacting a plant with any of the aforementioned compositions, and retrieving a progeny of the plant, wherein the progeny has an altered genome is provided. In certain embodiments, the contacting comprises phloem loading. In certain embodiments, the contacting with the composition occurs at the vegetative stage of the plant life cycle. Also provided are plants made by the method of producing a plant with an altered genome, comprising contacting a plant with any of the aforementioned compositions, and retrieving a progeny of the plant, wherein the progeny has an altered genome.

A meristem-delivery vector made up of a chimeric RNA having an RNA-guided nuclease component-containing segment and a meristem transport segment is provided.

A recombinant DNA having a sequence capable of producing as a transcript a meristem-delivery vector made up of a chimeric RNA having an RNA-guided nuclease component-containing segment and a meristem transport segment or an RNA that can be purified to form a composition comprising a substantially purified RNA molecule made up of a cargo segment fused to a meristem transport segment is provided.

Also provided are compositions comprising at least one RNA molecule comprising a cargo segment fused to a meristem transport segment (MTS), wherein the cargo segment comprises one or more guide RNAs for an RNA-guided nuclease. Use of the compositions to obtain a plant with an altered genome are provided.

Methods of producing a plant with an altered genome comprising (i) contacting a plant with at least a first composition comprising a cargo segment fused to a meristem transport segment (MTS), wherein the cargo segment comprises one or more guide RNAs for an RNA-guided nuclease; and (ii) retrieving a progeny of the plant, wherein the progeny has an altered genome, are provided. Plants comprising an altered genome made by the method are also provided.

BRIEF DESCRIPTION OF THE DRAWINGS/FIGURES

FIG. 1A, B is a diagram of the primary structure of an embodiment of an RNA sequence useful in methods for plant genomic alterations. g=guide RNA. In certain embodiments, the g or guide RNA segment may be made up of a spacer complementary to its genome target, and a crRNA,

which is part of the direct repeat sequences of Cas12a and/or Cas12j CRISPR arrays. The various labeled parts are not drawn to scale.

DETAILED DESCRIPTION

The phrase "allelic variant" as used herein refers to a polynucleotide or polypeptide sequence variant that occurs in a different strain, variety, or isolate of a given organism.

The term "and/or" where used herein is to be taken as specific disclosure of each of the two specified features or components with or without the other. Thus, the term "and/or" as used in a phrase such as "A and/or B" herein is intended to include "A and B," "A or B," "A" (alone), and "B" (alone). Likewise, the term "and/or" as used in a phrase such as "A, B, and/or C" is intended to encompass each of the following embodiments: A, B, and C; A, B, or C; A or C; A or B; B or C; A and C; A and B; B and C; A (alone); B (alone); and C (alone).

As used herein, the terms "Cas12a" and "Cpf1" are used interchangeably herein to refer to the same grouping of RNA directed nucleases.

As used herein, the terms "Cas12j" and "CasΦ" are used interchangeably herein to refer to the same grouping of RNA directed nucleases.

The term "fragment" refers to a contiguous set of polynucleotides or polypeptides. In one embodiment, a fragment is at least 10, 15, 20, or greater than 20 contiguous nucleotides. In other embodiments, a fragment is at least 10, 15, 20, or 50 to about 70, 90, 100, 120, 150, or 200 or more continuous nucleotides.

The term "isolated" as used herein means having been removed from its natural environment.

As used herein, the terms "include," "includes," and "including" are to be construed as at least having the features to which they refer while not excluding any additional unspecified features.

As used herein, the phrase "operably linked" or "fused" refers to a juxtaposition wherein the components so described are in a relationship permitting them to function in their intended manner. For instance, a promoter is operably linked to a coding sequence if the promoter affects its transcription or expression. In another non-limiting example, an RNA molecule comprising a "meristem transport sequence" (MTS) is operably linked or fused to a cargo RNA molecule if the MTS provides for delivery of the cargo RNA to meristem cells.

As used herein, the terms "orthologous" or "orthologue" are used to describe genes or the RNAs or proteins encoded by those genes that are from different species but which have the same function (e.g., encode RNAs which exhibit the same meristem transport function). Orthologous genes will typically encode RNAs or proteins with some degree of sequence identity (e.g., at least 40%, 50%, 60%, 70%, 80%, 90%, or 95% sequence identity) and can also exhibit conservation of sequence motifs, and/or conservation of structural features including RNA stem loop structures.

As used herein, the term "plant" includes a whole plant and any descendant, cell, tissue, or part of a plant. The term "plant parts" include any part(s) of a plant, including, for example and without limitation: seed (including mature seed and immature seed); a plant cutting; a plant cell; a plant cell culture; or a plant organ (e.g., pollen, embryos, flowers, fruits, shoots, leaves, roots, stems, and explants). A plant tissue or plant organ may be a seed, protoplast, callus, or any other group of plant cells that is organized into a structural or functional unit. A plant cell or tissue culture may be

capable of regenerating a plant having the physiological and morphological characteristics of the plant from which the cell or tissue was obtained, and of regenerating a plant having substantially the same genotype as the plant. Regenerable cells in a plant cell or tissue culture may be embryos, protoplasts, meristematic cells, callus, pollen, leaves, anthers, roots, root tips, silk, flowers, kernels, ears, cobs, husks, or stalks. In contrast, some plant cells are not capable of being regenerated to produce plants and are referred to herein as “non-regenerable” plant cells.

The phrase “substantially purified,” as used herein defines an isolation of a molecule or compound in a form that is substantially free of contaminants normally associated with the molecule or compound in a native or natural environment and means having been increased in purity as a result of being separated from other components of the original composition. The phrase “substantially purified RNA molecule” is used herein to describe an RNA molecule which has been separated from other contaminant compounds including, but not limited to polypeptides, lipids, and carbohydrates. In certain embodiments, a substantially purified RNA is at least 90%, 95%, 97%, 98%, 99%, 99.5%, or 99.9% free of contaminating compounds by weight. A substantially purified RNA molecule can be combined with other compounds including buffers, RNase inhibitors, surfactants, and the like in a composition.

To the extent to which any of the preceding definitions is inconsistent with definitions provided in any patent or non-patent reference incorporated herein by reference, any patent or non-patent reference cited herein, or in any patent or non-patent reference found elsewhere, it is understood that the preceding definition will be used herein.

The reagents and methods described provide a relatively easy and convenient solution for producing plants with altered genomes, i.e. individuals with designed mutations (i.e., DNA sequence changes including insertions, deletions, and substitutions (Indels)). In most embodiments, the methods and systems rely on RNA molecules produced with established molecular biology techniques. The RNA molecules, which comprise genome-editing reagents, are then introduced into a plant and taken up into meristematic cells. The meristematic cell genomes are thus altered, and the mutations (i.e., DNA sequence changes including Indels) are carried into germline cells and subsequent generations.

Meristem transport segments travel through the plant, typically via the phloem, and are taken up into meristematic tissues. The examples below are sequences from individual species, which sometimes work across species. For example, Arabidopsis FT-based vectors work in *Nicotiana benthamiana* and Arabidopsis. But, vectors can be designed based on alternative sequences, which can be based either on the species subject to genomic editing, or based on a closely related species.

While the transport segment is based on a plant-transported RNA, its actual sequence may be a fragment determined by characterizing a deletion series to make a smaller sequence retaining the desired transport (phloem mobility and/or meristem cell translocation) capabilities. In certain embodiments, the meristem transport segment is a sub-fragment of a plant transported RNA identified by assaying a deletion series for a smaller sequence retaining the desired transport (phloem mobility and/or meristem cell translocation) function. The initiator methionine codon or translation initiation codon of the base sequence may also be mutated in some cases.

The flowering time (FT) mRNA is useful as a meristem transport segment. SEQ ID NO: 2 shows the DNA sequence

that encodes the Arabidopsis FT RNA, and SEQ ID NO: 1 is a fraction of SEQ ID NO: 2 that encodes the RNA that functions as a transport segment. Alternative useful FTs may be ZCN8 (encoded by SEQ ID NO: 3), which may work across related monocot species. Alternative useful FTs may be GmFT2a (Sun et al. PLoS One. 2011; 6(12):e29238. doi:10.1371/journal.pone.0029238; Jiang et al. BMC Genomics. 2019; 20(1):230. doi: 10.1186/s12864-019-5577-5; Kong et al. Plant Physiol. 2010 November; 154(3): 1220-31. doi: 10.1104/pp.110.160796; Takeshima et al. J Exp Bot. 2019 Aug. 7; 70(15):3941-3953. doi: 10.1093/jxb/erz199), which may work across related dicot species. FT RNA molecules that can be used include: (i) RNAs set forth in SEQ ID NO: 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, or 24, a meristem transport-competent (MTC) ortholog thereof, a MTC variant thereof, and/or a MTC fragment thereof; (ii) allelic variants of SEQ ID NO: 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, or 24, a meristem transport-competent (MTC) ortholog thereof, a MTC variant thereof, and/or a MTC fragment thereof; and (iii) FT RNAs from various plants set forth in U.S. 20190300890, which is incorporated herein by reference in its entirety, allelic variants thereof, and meristem transport-competent (MTC) orthologs thereof, MTC variants thereof, and/or MTC fragments thereof. FT RNA molecules that can be used include RNAs having at least 85%, 90%, 95%, 98%, or 99% sequence identity to SEQ ID NO: 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, or a meristem transport-competent (MTC) fragment thereof;

More generally, viral and cellular-derived RNA molecules that are useful as part of a transport segment include the mRNAs of FT, GAI, CmNACP, LeT6 a tomato KNOX gene, BEL5, or tRNA-like sequences (Ruiz-Medrano et al., 1999 Phloem long-distance transport of CmNACP mRNA: implications for supracellular regulation in plants. Development 126, 4405-4419; Kim et al., 2001 Developmental changes due to long-distance movement of a homeobox fusion transcript in tomato. Science 293, 287-289; Haywood et al., 2005 Phloem long distance trafficking of GIBBERELLIC ACID-INSENSITIVE RNA regulates leaf development. Plant J. 42, 49-68; and Li et al., 2011 Mobile FT mRNA contributes to the systemic florigen signaling in floral induction. Sci. Rep. 1, 73; Cho et al., 2015, J. Exp. Bot. 66: 6835-6847; Zhang et al., 2016, Plant Cell, 28: 1237-1249; and WO2017178633). GAI RNAs that can be used include: (i) RNAs set forth in SEQ ID NO: 26, a meristem transport-competent (MTC) ortholog thereof, a MTC variant thereof, and/or a MTC fragment thereof; (ii) allelic variants of SEQ ID NO: 26, a meristem transport-competent (MTC) ortholog thereof, a MTC variant thereof, and/or a MTC fragment thereof; and (iii) RNAs having at least 85%, 90%, 95%, 98%, or 99% sequence identity to SEQ ID NO: 26, or a meristem transport-competent (MTC) fragment thereof. CmNACP RNAs that can be used include: (i) RNAs set forth in SEQ ID NO: 25, a meristem transport-competent (MTC) ortholog thereof, a MTC variant thereof, and/or a MTC fragment thereof; (ii) allelic variants of SEQ ID NO: 25, a MTC variant thereof, and/or a MTC fragment thereof; and (iii) RNAs having at least 85%, 90%, 95%, 98%, or 99% sequence identity to SEQ ID NO: 25, or a meristem transport-competent (MTC) fragment thereof. LeT6 RNAs that can be used include: (i) RNAs set forth in SEQ ID NO: 27, a meristem transport-competent (MTC) ortholog thereof, a MTC variant thereof, and/or a MTC fragment thereof; (ii) allelic variants of SEQ ID NO: 27, a MTC variant thereof, and/or a MTC fragment thereof; and (iii) RNAs having at

least 85%, 90%, 95%, 98%, or 99% sequence identity to SEQ ID NO: 27, or a meristem transport-competent (MTC) fragment thereof. BEL5 RNAs that can be used include: (i) RNAs set forth in SEQ ID NO: 28, a meristem transport-competent (MTC) ortholog thereof, a MTC variant thereof, and/or a MTC fragment thereof; (ii) allelic variants of SEQ ID NO: 28, a MTC variant thereof, and/or a MTC fragment thereof; and (iii) RNAs having at least 85%, 90%, 95%, 98%, or 99% sequence identity to SEQ ID NO: 28, or a meristem transport-competent (MTC) fragment thereof. Examples of tRNA-like RNAs that can be used include: (i) RNAs set forth in SEQ ID NO: 29, 30, a meristem transport-competent (MTC) ortholog thereof, a MTC variant thereof, and/or a MTC fragment thereof; (ii) allelic variants of SEQ ID NO: 29, 30, a MTC variant thereof, and/or a MTC

fragment thereof, and (iii) RNAs having at least 85%, 90%, 95%, 98%, or 99% sequence identity to SEQ ID NO: 29, 30, or a meristem transport-competent (MTC) fragment thereof. In certain embodiments, a TLS sequence, SEQ ID NO: 29 or 30, a meristem transport-competent (MTC) ortholog thereof, a MTC variant thereof, and/or an MTC fragment thereof can comprise an RNA hairpin comprising a first stem of 8 to 12 nucleotides, at least one variable bulge, a second stem of 4 to 7 nucleotides, and a variable loop. TLS sequences suitable for RNA transport and the structural features of such RNAs are set forth in Zhang et al. Plant Cell. 2016 June; 28(6): 1237, doi.org/10.1105/tpc.15.01056.

Further description of biological sequences provided in the sequence listing is set forth in Table 1. RNA molecules set forth in SEQ ID NO: 9-30 are respectively encoded by the DNA molecules set forth in SEQ ID NO: 31-52.

TABLE 1

Description of biological sequences.		
SEQ ID NO:	TYPE	Comments
1	DNA	<i>Arabidopsis thaliana</i>
2	DNA	NM_001334207.1 <i>Arabidopsis thaliana</i> PEBP (phosphatidylethanolamine-binding protein) family protein (FT), mRNA
3	DNA	EU241924.1 <i>Zea mays</i> ZCN8 (ZCN8) mRNA, complete cds
4	DNA	GmFT2a CDS, the soy FT ortholog according to Sun et al., 2011 and Cai et al., 2018 (GenBank ID: EU287455)
5	RNA	RNA encoded by SEQ ID NO: 1
6	RNA	RNA encoded by SEQ ID NO: 2
7	RNA	RNA encoded by SEQ ID NO: 3
8	RNA	RNA encoded by SEQ ID NO: 4
9	RNA	DQ865290.1 <i>Cucurbita maxima</i> flowering locus T-like 1 (FTL1) mRNA, complete cds
10	RNA	DQ865291.1 <i>Cucurbita maxima</i> flowering locus T-like 2 (FTL2) mRNA, complete cds
11	RNA	DQ871590.1 <i>Vitis vinifera</i> FT-like protein (FT) mRNA, complete cds
12	RNA	AB161112.1 <i>Malus x domestica</i> MdFT1 mRNA for flowering locus T like protein, complete cds
13	RNA	AB027456.1 <i>Citrus unshiu</i> CiFT mRNA, complete cds
14	RNA	AY186735.1: 2002-2199, 2287-2348, 4490-4530, 5586-5818 <i>Lycopersicon esculentum</i> SP3D (SP3D) gene, complete cds
15	RNA	DQ387859.1 <i>Populus tremula</i> flowering locus T-like protein FT1 (FT1) mRNA, complete cds
16	RNA	>DQ100327.1: 1332-1532, 1950-2011, 2121-2391 <i>Hordeum vulgare</i> subsp. <i>vulgare</i> FT-like protein (FT1) gene, complete cds
17	RNA	DQ297407.1: 955-1164, 1235-1296, 3672-3712, 3808-4031 <i>Hordeum vulgare</i> subsp. <i>vulgare</i> FT-like protein (FT2) gene, complete cds
18	RNA	AB052944.1 <i>Oryza sativa</i> Japonica Group Hd3a mRNA, complete cds, cultivar: Nipponbare
19	RNA	AB062676.1 <i>Oryza sativa</i> Japonica Group RFT1 mRNA for FT-like protein, complete cds
20	RNA	EU178859.1 <i>Ipomoea nil</i> FT-like protein (FT1) mRNA, complete cds
21	RNA	AB027506.1 <i>Arabidopsis thaliana</i> TSF (TWIN SISTER OF FT) mRNA, complete cds
22	RNA	LC128590.1: 3049-3243, 3377-3438, 3830-3870, 4102-4322 <i>Glycine max</i> FT5a gene for flowering locus T, complete cds, cultivar: Toyoharuka
23	RNA	ZmZCN9 NM_001112777.2 <i>Zea mays</i> ZCN9 protein (LOC100127520), mRNA
24	RNA	ZmZCN10
25	RNA	>EU241926.1 <i>Zea mays</i> ZCN10 (ZCN10) mRNA, complete cds
26	RNA	CmNACP: >FJ151402.1 <i>Cucurbita maxima</i> NAC-domain containing protein (NACP1) mRNA, complete cds
27	RNA	GAI: >Y15193.1 <i>Arabidopsis thaliana</i> GAI gene
28	RNA	LeT6 a tomato KNOX gene: >AF000141.1 <i>Lycopersicon esculentum</i> class I knotted-like homeodomain protein (LeT6) mRNA, complete cds

TABLE 1-continued

Description of biological sequences.		
SEQ ID NO:	TYPE	Comments
28	RNA	BEL5: >NM_001287992.1 <i>Solanum tuberosum</i> BEL1-related homeotic protein 5 (BEL5), mRNA
29	RNA	AT5G57885.1 (tRNA-Met)
30	RNA	AT1G71700 (tRNA-Gly)
31	DNA	DQ865290.1 <i>Cucurbita maxima</i> flowering locus T-like 1 (FTL1) mRNA, complete cds
32	DNA	DQ865291.1 <i>Cucurbita maxima</i> flowering locus T-like 2 (FTL2) mRNA, complete cds
33	DNA	DQ871590.1 <i>Vitis vinifera</i> FT-like protein (FT) mRNA, complete cds
34	DNA	AB161112.1 <i>Malus x domestica</i> MdFT1 mRNA for flowering locus T like protein, complete cds
35	DNA	AB027456.1 <i>Citrus unshiu</i> CiFT mRNA, complete cds
36	DNA	AY186735.1: 2002-2199, 2287-2348, 4490-4530, 5586-5818 <i>Lycopersicon esculentum</i> SP3D (SP3D) gene, complete cds
37	DNA	DQ387859.1 <i>Populus tremula</i> flowering locus T-like protein FT1 (FT1) mRNA, complete cds
38	DNA	>DQ100327.1: 1332-1532, 1950-2011, 2121-2391 <i>Hordeum vulgare</i> subsp. <i>vulgare</i> FT-like protein (FT1) gene, complete cds
39	DNA	DQ297407.1: 955-1164, 1235-1296, 3672-3712, 3808-4031 <i>Hordeum vulgare</i> subsp. <i>vulgare</i> FT-like protein (FT2) gene, complete cds
40	DNA	AB052944.1 <i>Oryza sativa</i> Japonica Group Hd3a mRNA, complete cds, cultivar: Nipponbare
41	DNA	AB062676.1 <i>Oryza sativa</i> Japonica Group RFT1 mRNA for FT-like protein, complete cds
42	DNA	EU178859.1 <i>Ipomoea nil</i> FT-like protein (FT1) mRNA, complete cds
43	DNA	AB027506.1 <i>Arabidopsis thaliana</i> TSF (TWIN SISTER OF FT) mRNA, complete cds
44	DNA	LC128590.1: 3049-3243, 3377-3438, 3830-3870, 4102-4322 <i>Glycine max</i> FT5a gene for flowering locus T, complete cds, cultivar: Toyoharuka
45	DNA	ZmZCN9 NM_001112777.2 <i>Zea mays</i> ZCN9 protein (LOC100127520), mRNA
46	DNA	ZmZCN10 >EU241926.1 <i>Zea mays</i> ZCN10 (ZCN10) mRNA, complete cds
47	DNA	CmNACP: >FJ151402.1 <i>Cucurbita maxima</i> NAC-domain containing protein (NACP1) mRNA, complete cds
48	DNA	GAI: >Y15193.1 <i>Arabidopsis thaliana</i> GAI gene
49	DNA	LeT6 a tomato KNOX gene: >AF000141.1 <i>Lycopersicon esculentum</i> class I knotted-like homeodomain protein (LeT6) mRNA, complete cds
50	DNA	BEL5: >NM_001287992.1 <i>Solanum tuberosum</i> BEL1-related homeotic protein 5 (BEL5), mRNA
51	DNA	AT5G57885.1 (tRNA-Met)
52	DNA	AT1G71700 (tRNA-Gly)
53	PRO	FnCas12a (UniProtKB/Swiss-Prot: A0Q7Q2.1); US20160208243; and WO 2017/189308)
54	RNA	FnCas12aDR (Fonfara et al. Nature 532, 517-521 (2016). doi.org/10.1038/nature17945; US2016-0208243; WO 2017/189308)
55	PRO	LbCpfI (from Lachnospiraceae bacterium ND2006; UniProtKB: A0A182DWE3)
56	RNA	LbCpfI DR (from Lachnospiraceae bacterium ND2006; Zetsche et al., doi.org/10.1101/134015)
57	PRO	Cas12j-1 protein (Pausch et al., 2020 Science 17 Jul. 2020: Vol. 369, Issue 6501, pp. 333-337)
58	RNA	Cas12j-2 DR sequence (Pausch et al., 2020 Science 17 Jul. 2020: Vol. 369, Issue 6501, pp. 333-337)
59	PRO	Cas12j-2 protein (Pausch et al., 2020 Science 17 Jul. 2020: Vol. 369, Issue 6501, pp. 333-337)
60	RNA	Cas12j-2 DR sequence (Pausch et al., 2020 Science 17 Jul. 2020: Vol. 369, Issue 6501, pp. 333-337)
61	PRO	Cas12j-3 protein (Pausch et al., 2020 Science 17 Jul. 2020: Vol. 369, Issue 6501, pp. 333-337)
62	RNA	Cas12j-3 DR sequence (Pausch et al., 2020 Science 17 Jul. 2020: Vol. 369, Issue 6501, pp. 333-337)

The meristem transport-competence (MTC) potential can be determined for any variants, fragments, and/or orthologs of the aforementioned FT, GAI, CmNACP, LeT6 a tomato KNOX gene, BEL5, or tRNA-like RNAs. A side-by-side comparison with a known MTS as a positive control is useful. As such, a number of configurations can be used. One approach is to fuse candidate sequences to guide sequences of characterized editing potential for a species of interest. RNA sequences can be introduced into the phloem of an individual plant that expresses at least in the meristem a nuclease capable of associating with the guide sequence and producing the intended genomic alteration. The RNA sequences can be expressed *in vitro*, and introduced into the phloem as purified molecules. For example, a concentrated solution of RNA molecules of interest can be applied to a mechanically injured plant tissue, such as a cut or abraded leaf, stem, or meristem dome. RNAs can be coated on particles, such as micro or nano-scale particles such as gold or tungsten, for biolistic delivery. Alternatively, the RNA sequences could be incorporated into RNA viruses introduced in the plants (Jackson et al. 2012, *Front. Plant Sci.* 3, 127; Ali et al. 2015, *Mol. Plant* 8, 1288-1291; Cody et al. 2017 *Plant Physiol.* 175, 23-35; Ali et al. 2018, *Virus Res.* 244, 333-337; Gao et al. 2019, *New Phytol.* 223, 2120-2133), or the MTC can be assayed by introducing RNAs by grafting, i.e. the RNA molecules can be expressed in the rootstock of a grafted plant, and their effect observed in the scion (Zhang et al., 2016, *Plant Cell*, 28: 1237-1249; Huang et al, 2018, *Plant Physiol.* 178:783-794). MTS candidates can be assayed for longer and/or more complex RNA molecules, or mixtures of RNA molecules, that comprise not only guide or processable guide regions, but also nuclease-encoding sequences.

A clear readout of MTC is detection of the expected genomic alterations in progeny plants, which can be done by sequencing of the target genomic region, or even by whole genome sequencing. But alternative readouts can be designed that may be more convenient in some cases. For example, the guide sequences may be directed to disrupt or repair a reporter gene, such as a transgene encoding a fluorescent polypeptide. The expected genetic changes can then be evaluated in the treated plants by measuring changes in the reporter. Another convenient genomic alteration target in many species is phytoene desaturase (PDS), with the albino phenotype serving as a readout.

The cargo segments of the engineered RNA deliver the genome-editing components. In general, these will be based on CRISPR-Cas systems, but some alternatives are possible. The alternatives include RNAi for heritable knock-down as affected by DNA methylation status, a TALEN, a zinc finger nucleases (ZFN), and a meganuclease.

In certain embodiments, an RNA molecule comprising a RNA segment encoding a ZFN (e.g., a zinc finger nuclease or zinc finger nickase) that is operably linked to an RNA segment comprising an MTS to provide for ZFN-mediated gene editing in a plant meristem. Zinc-finger nucleases are site-specific endonucleases comprising two protein domains: a DNA-binding domain, comprising a plurality of individual zinc finger repeats that each recognize between 9 and 18 base pairs, and a DNA-cleavage domain that comprises a nuclease domain (typically FokI). The cleavage domain dimerizes in order to cleave DNA; therefore, a pair of ZFNs are required to target non-palindromic target polynucleotides. In certain embodiments, zinc finger nuclease and zinc finger nickase design methods which have been described (Urnov et al. (2010) *Nature Rev. Genet.*, 11:636-646; Mohanta et al. (2017) *Genes* vol. 8,12: 399; Ramirez et

al. *Nucleic Acids Res.* (2012); 40(12): 5560-5568; Liu et al. (2013) *Nature Communications*, 4: 2565) can be adapted for use in the methods set forth herein. The zinc finger binding domains of the zinc finger nuclease or nickase provide specificity and can be engineered to specifically recognize any desired target DNA sequence. The zinc finger DNA binding domains are derived from the DNA-binding domain of a large class of eukaryotic transcription factors called zinc finger proteins (ZFPs). The DNA-binding domain of ZFPs typically contains a tandem array of at least three zinc "fingers" each recognizing a specific triplet of DNA. A number of strategies can be used to design the binding specificity of the zinc finger binding domain. One approach, termed "modular assembly", relies on the functional autonomy of individual zinc fingers with DNA. In this approach, a given sequence is targeted by identifying zinc fingers for each component triplet in the sequence and linking them into a multifinger peptide. Several alternative strategies for designing zinc finger DNA binding domains have also been developed. These methods are designed to accommodate the ability of zinc fingers to contact neighboring fingers as well as nucleotide bases outside their target triplet. Typically, the engineered zinc finger DNA binding domain has a novel binding specificity, compared to a naturally-occurring zinc finger protein. Engineering methods include, for example, rational design and various types of selection. Rational design includes, for example, the use of databases of triplet (or quadruplet) nucleotide sequences and individual zinc finger amino acid sequences, in which each triplet or quadruplet nucleotide sequence is associated with one or more amino acid sequences of zinc fingers which bind the particular triplet or quadruplet sequence. See, e.g., U.S. Pat. Nos. 6,453,242 and 6,534,261, both incorporated herein by reference in their entirety. Exemplary selection methods (e.g., phage display and yeast two-hybrid systems) can be adapted for use in the methods described herein. In addition, enhancement of binding specificity for zinc finger binding domains has been described in U.S. Pat. No. 6,794,136, incorporated herein by reference in its entirety. In addition, individual zinc finger domains may be linked together using any suitable linker sequences. Examples of linker sequences are publicly known, e.g., see U.S. Pat. Nos. 6,479,626; 6,903,185; and 7,153,949, incorporated herein by reference in their entirety. The nucleic acid cleavage domain is non-specific and is typically a restriction endonuclease, such as FokI. This endonuclease must dimerize to cleave DNA. Thus, cleavage by FokI as part of a ZFN requires two adjacent and independent binding events, which must occur in both the correct orientation and with appropriate spacing to permit dimer formation. The requirement for two DNA binding events enables more specific targeting of long and potentially unique recognition sites. FokI variants with enhanced activities have been described and can be adapted for use in the methods described herein; see, e.g., Guo et al. (2010) *J. Mol. Biol.*, 400:96-107.

In certain embodiments, an RNA molecule comprising a RNA segment encoding a TALEN (e.g., a TALE nuclease or nickase) that is operably linked to an RNA segment comprising an MTS to provide for TALEN-mediated gene editing in a plant meristem. Transcription activator like effectors (TALEs) are proteins secreted by certain *Xanthomonas* species to modulate gene expression in host plants and to facilitate the colonization by and survival of the bacterium. TALEs act as transcription factors and modulate expression of resistance genes in the plants. Recent studies of TALEs have revealed the code linking the repetitive region of TALEs with their target DNA-binding sites.

TALEs comprise a highly conserved and repetitive region consisting of tandem repeats of mostly 33 or 34 amino acid segments. The repeat monomers differ from each other mainly at amino acid positions 12 and 13. A strong correlation between unique pairs of amino acids at positions 12 and 13 and the corresponding nucleotide in the TALE-binding site has been found. The simple relationship between amino acid sequence and DNA recognition of the TALE binding domain allows for the design of DNA binding domains of any desired specificity. TALEs can be linked to a non-specific DNA cleavage domain to prepare genome editing proteins, referred to as TAL-effector nucleases or TALENs. As in the case of ZFNs, a restriction endonuclease, such as FokI, can be conveniently used. Methods for use of TALENs in plants have been described and can be adapted for use in the methods described herein, see Mahfouz et al. (2011) *Proc. Natl. Acad. Sci. USA*, 108:2623-2628; Mahfouz (2011) *GM Crops*, 2:99-103; and Mohanta et al. (2017) *Genes* vol. 8, 12: 399). TALE nickases have also been described and can be adapted for use in methods described herein (Wu et al.; *Biochem Biophys Res Commun.* (2014); 446(1):261-6; Luo et al; *Scientific Reports* 6, Article No.: 20657 (2016)).

Plants comprising the RNA molecules that comprise cargo segments that are operably linked to MTS sequences are also provided herein. In certain embodiments, such RNA molecules will be present at detectable concentrations in the plants for only a certain period of time following. For example, the concentrations of RNA molecules comprising guide RNAs separated by processing elements comprising direct repeats (DR, i.e., pre-crRNAs comprising a full-length direct repeat (full-DR-crRNA)) which are capable of being processed (i.e., cleaved) by an RNA-guided nuclease are expected to decrease over time when the RNA-guided nuclease is also present in the plant. The concentrations of RNA molecules comprising guide RNAs separated by processing elements comprising direct repeats which are capable of being processed by an RNA-guided nuclease are also expected to be decreased in tissues where the RNA-guided nuclease is located. Nonetheless, the unprocessed RNA molecules can be detected by a variety of techniques that include reverse transcriptase PCR (RT-PCR) assays where oligonucleotide primers and optionally detection probes which specifically amplify and detect the unprocessed RNA molecule comprising the cargo segments that are operably linked to MTS sequences are used. Such plants can comprise any of the RNA molecules or combinations of RNA molecules present in the compositions provided herein that are used to contact the plants. In certain embodiments, an active form of the RNA guided nuclease is predominantly localized in meristem tissue of the plant. In certain embodiments, the RNA-guided nuclease can be encoded by an RNA molecule that is optionally further comprises an operably linked MTS sequence. In certain embodiments, the RNA-guided nuclease can be encoded by DNA that is operably linked to promoters that include a meristem-preferred or meristem-specific promoter which is active in meristem cells. DNA encoding the RNA-guided nuclease can be provided in a transgene that is stably integrated in the genome of the plant, in DNA that is not integrated into the plant genome, or in DNA provided in a viral vector (e.g., a geminivirus replicon). Geminivirus DNA replicons suitable for delivery of DNA molecules encoding an RNA-guided nuclease to plants include a Beet Yellow Dwarf Virus replicon (Baltes, Nicholas J. et al. *Plant Cell* vol. 26, 1 (2014): 151-63. doi:10.1105/tpc.113.119792).

It is understood that for all systems, the use of a nuclease activity for cutting DNA followed by repair by the endogenous cell machinery is one solution to generate useful mutants. The nuclease activity can be eliminated or altered, as in dCas or nCas, TALE or ZF versions of the polypeptides. The inactivated nucleases can be useful for targeting the desired DNA sequence, while editing can be performed by nucleobase editors attached to the altered nucleases. Examples are included in WO2018176009 and U.S. Pat. No. 10,113,163, incorporated herein by reference.

CRISPR-based RNA-guided nuclease systems typically require an effector polypeptide, and one or more guide RNAs. The guide RNAs are generally made up of an effector-binding region and a target DNA recognition region, and in some embodiments include tracrRNAs. Useful CRISPR-based RNA-guided nuclease systems have been described and are known from the literature as Cas9, Cas12a (Cpf1), Cas12e (CasX), Cas12d (CasY), C2c1, C2c2, and C2c3, (see WO2018176009) Cas12h, Cas12i (see Yan et al. 2019, *Science* Vol. 363, Issue 6422, pp. 88-91) and Cas12j (Pausch et al., 2020 *Science* 17 Jul. 2020: Vol. 369, Issue 6501, pp. 333-337).

The Cas nuclease or effector polypeptide is intended to be translated inside a plant meristem cell. As such, it is typically embedded within an mRNA component. A 5' cap and polyA tail are also useful in stabilizing the RNA. A 5' UTR has translation initiation sequences upstream of the Cas coding sequence. For example, an mRNA can comprise a 5'UTR comprising a 7-methylguanosine cap at its 5' terminus followed by an untranslated sequence and terminated by the translation initiation codon of the coding sequence (e.g., the CAS coding sequence).

Cargo containing guide RNA can be part of the same RNA (mRNA) capable of expressing the Cas nuclease. In one embodiment, one or more guide RNAs are flanked by direct repeats (DR) of the CRISPR array from which the Cas effector polypeptide was first isolated. For example, a translated and expressed active Cas12a nuclease can process the DR-flanked spacers of the cargo RNA to make guide RNAs. In certain embodiments, a translated and expressed active Cas12j nuclease can process Cas12j DR-flanked spacers of the cargo RNA to make guide RNAs. Alternatively, guide RNA suitable for matching expressed effector polypeptide can be flanked by processing elements, so that functional guide RNAs are excised inside the cells. Exemplary processing elements include hammerhead ribozymes, Csy4, and tRNAs (see Mikami et al, *Plant Cell Physiol.* 2017 November; 58(11): 1857-1867, and U.S. Pat. No. 10,308,947).

In certain embodiments, an MTS is operably linked to a cargo segment comprising an array of a plurality of guide RNAs (e.g., 2, 3, 4, or more guide RNAs) separated by processing elements to provide for gene editing at a plurality of genomic locations targeted by each guide RNA. In certain embodiments, the plurality of guide RNAs are separated by processing elements comprising direct repeats (DR; i.e., pre-crRNAs comprising a full-length direct repeat (full-DR-crRNA)) which are capable of being processed (i.e., cleaved) by an RNA-guided nuclease. Examples of such DRs include the Cas12a DR (e.g., SEQ ID NO: 54 or 56) which can be cleaved by a Cas12a guided nuclease (e.g., SEQ ID NO: 53 or 55, respectively). Cleavage of RNAs comprising Cas12a DRs by Cas12a has been described (Fonfara et al. *Nature* 532, 517-521 (2016). doi.org/10.1038/nature17945; U.S. 20160208243; WO 2017/189308). Other examples of such DRs include the Cas12j DRs (e.g., SEQ ID NO: 58, 60, or 62) which can be cleaved by a Cas12j guided nuclease (e.g., SEQ ID NO: 57, 59, or 61, respectively). In

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such embodiments, the crRNA portion of the DR can remain as a part of the gRNA after processing and can be recognized by the RNA guided nuclease to provide for editing of genomic DNA recognized via hybridization of the gRNA to the targeted genomic site.

Compositions comprising: (i) RNA molecules comprising an MTS is operably linked to a cargo segment; (ii) nucleic acids encoding RNA guided nucleases; and/or (iii) donor DNA templates can further comprise components that include:

- (a) solvents (e.g., water, dimethylsulfoxide, dimethylformamide, acetonitrile, N-pyrrolidine, pyridine, hexamethylphosphoramide, alcohols, alkanes, alkenes, dioxanes, polyethylene glycol, and other solvents miscible or emulsifiable with water or that will dissolve phosphonucleotides in non-aqueous systems);
- (b) fluorocarbons (e.g., perfluorodecalin, perfluoromethyldecalin);
- (c) glycols or polyols (e.g., propylene glycol, polyethylene glycol);
- (d) surfactants, including cationic surfactants, anionic surfactants, non-ionic surfactants, and amphiphilic surfactants, e.g., alkyl or aryl sulfates, phosphates, sulfonates, or carboxylates; primary, secondary, or tertiary amines; quaternary ammonium salts; sultaines, betaines; cationic lipids; phospholipids; tallowamine; bile acids such as cholic acid; saponins or glycosylated triterpenoids or glycosylated sterols (e.g., saponin commercially available as catalogue number 47036-50 g-F, Sigma-Aldrich, St. Louis, MO); long chain alcohols; organosilicone surfactants including nonionic organosilicone surfactants such as trisiloxane ethoxylate surfactants or a silicone polyether copolymer such as a copolymer of polyalkylene oxide modified heptamethyl trisiloxane and allyloxypolypropylene glycol methylether (commercially available as SILWET L-77™ brand surfactant having CAS No. 27306-78-1 and EPA Number CAL. REG. No. 5905-50073-AA, Momentive Performance Materials, Inc., Albany, N.Y.); specific examples of useful surfactants include sodium lauryl sulfate, the Tween series of surfactants, Triton-X100, Triton-X114, CHAPS and CHAPSO, Tergitol-type NP-40, Nonidet P-40;
- (e) lipids, lipoproteins, lipopolysaccharides;
- (f) acids, bases, caustic agents; buffers;
- (g) peptides, proteins, or enzymes (e.g., cellulase, pectolyase, maceroenzyme, pectinase), including cell-penetrating or pore-forming peptides (e.g., (BO100)2K8, Genscript; poly-lysine, poly-arginine, or poly-homoarginine peptides; gamma zein, see U.S. Patent Application publication 2011/0247100, incorporated herein by reference in its entirety; transcription activator of human immunodeficiency virus type 1 ("HIV-1 Tat") and other Tat proteins, see, e.g., [www\[dot\]lifetein\[dot\]com/Cell_Penetrating_Peptides\[dot\]html](http://www.lifetein.com/Cell_Penetrating_Peptides.html) and Järver (2012) *Mol. Therapy-Nucleic Acids*, 1:e27, 1-17); octa-arginine or nona-arginine; poly-homoarginine (see Unnamalai et al. (2004) *FEBS Letters*, 566: 307-310); see also the database of cell-penetrating peptides CPPsite 2.0 publicly available at [crdd\[dot\]osdd\[dot\]net/raghava/cppsite/](http://crdd[dot]osdd[dot]net/raghava/cppsite/);
- (h) RNase inhibitors;
- (i) cationic branched or linear polymers such as chitosan, poly-lysine, DEAE-dextran, polyvinylpyrrolidone ("PVP"), or polyethylenimine ("PEI", e.g., PEI,

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branched, MW 25,000, CAS #9002-98-6; PEI, linear, MW 5000, CAS #9002-98-6; PEI linear, MW 2500, CAS #9002-98-6);

- (j) dendrimers (see, e.g., U.S. Patent Application Publication 2011/0093982, incorporated herein by reference in its entirety);
- (k) counter-ions, amines or polyamines (e.g., spermine, spermidine, putrescine), osmolytes, buffers, and salts (e.g., calcium phosphate, ammonium phosphate);
- (l) polynucleotides (e.g., non-specific double-stranded DNA, salmon sperm DNA);
- (m) transfection agents (e.g., Lipofectin®, Lipofectamine®, and Oligofectamine®, and InvivoFectamine® (all from Thermo Fisher Scientific, Waltham, MA), PepFect (see Ezzat et al. (2011) *Nucleic Acids Res.*, 39:5284-5298), TransIt® transfection reagents (Mirus Bio, LLC, Madison, WI), and poly-lysine, poly-homoarginine, and poly-arginine molecules including octo-arginine and nono-arginine as described in Lu et al. (2010) *J. Agric. Food Chem.*, 58:2288-2294);
- (n) antibiotics, including non-specific DNA double-strand-break-inducing agents (e.g., phleomycin, bleomycin, talisomycin);
- (o) antioxidants (e.g., glutathione, dithiothreitol, ascorbate); and/or
- (p) chelating agents (e.g., EDTA, EGTA).

Compositions comprising: (i) RNA molecules comprising an MTS is operably linked to a cargo segment; (ii) nucleic acids encoding RNA guided nucleases; and/or (iii) donor DNA templates can be delivered to the plant and/or meristem cells of the plant by particle mediated delivery, and any other direct method of delivery, such as but not limiting to, Agrobacterium-mediated transformation, polyethylene glycol (PEG)-mediated transfection to protoplasts, whiskers mediated transformation, electroporation, particle bombardment, and/or by use of cell-penetrating peptides.

In certain embodiments, plants are contacted either simultaneously or sequentially with one, two, three or more RNA molecules in one or more compositions where at least one of the RNA molecules comprises an MTS operably linked to a cargo segment comprising at least one guide RNA. In certain embodiments, one of the RNA molecules comprises an MTS operably linked to a cargo segment comprising at least one guide RNA and the other RNA molecule encoding an RNA guided nuclease and optionally an MTS, where the RNA guided nuclease can process the RNA comprising the guide RNA to release a functional guide RNA. In certain embodiments, one of the RNA molecules comprises an MTS operably linked to a cargo segment comprising at least one guide RNA and the other RNA molecule comprises an RNA guided nuclease and optionally an MTS, where the RNA guided nuclease cannot process the RNA comprising the guide RNA to release a functional guide RNA (e.g., processing elements present in the RNA molecule comprising the gRNA and the MTS are not recognized by the RNA-guided nuclease). In certain embodiments, guide RNAs of the first and second RNA molecule are flanked by or comprise processing elements (e.g., DRs) which are processed by different RNA-guided nuclease (e.g., a Cas12a nuclease can process the first RNA molecule and a Cas12j nuclease can process the second RNA molecule). In certain embodiments, the cargo segment of the first RNA molecule comprises guide RNAs which are distinct from the guide RNAs of the cargo segment second RNA molecule. Such distinct gRNAs provided by the first RNA molecule can provide for genome editing at one or more first genomic sites

in a meristem cell while the distinct gRNAs provided by the second RNA molecule can provide for genome editing at one or more second genomic sites in a meristem cell. Such contacting the plant with RNA molecules in a composition can occur sequentially such that the first gRNA(s) are delivered, allowed sufficient time (e.g., about 6, 12, 18 or 20 to about 24, 30, or 36 hours) to effect desired genome edits, followed by contacting the plant with the second RNA molecules in a second composition to deliver the second gRNA(s) to effect additional desired genome edits, where such desired genome edits are effected by providing the gRNA(s) and an RNA guided nuclease in at least the meristem cell. Without seeking to be limited by theory, it is believed that cutting chromosomes at multiple location simultaneously is cytotoxic and that such cytotoxicity can be mitigated by delivering a limited number of guide RNAs at different times (e.g., about 6, 12, 18 or 20 to about 24, 30, or 36 hours apart). In certain embodiments, a plant can be contacted by one or more RNA molecules that comprise at least one gRNA operably linked to an MTS, optionally along with an RNA encoding RNA guided nuclease, permitted a sufficient period of time to accumulate the RNA molecule in the meristem cells (e.g., about 6, 12, 18 or 20 to about 24, 30, or 36 hours apart), and then contacted with a different mixture of one or more RNA molecules that comprise at least one different gRNA operably linked to an MTS, optionally along with an RNA encoding RNA guided nuclease, where the RNA guided nuclease can process the RNA comprising the guide RNA to release a functional guide RNA and/or effect a desired genomic edit with the gRNA in the meristem cells.

In certain embodiments, the RNA molecules comprising at least one gRNA fused to an MTS are provided in combination with the RNA guided nuclease and a donor DNA template to effect insertions of DNA elements in the donor DNA template at the target editing site in the plant genome by homology dependent repair (HDR), non-homologous end joining (NHEJ), or microhomology-mediated end joining (MMEJ). Donor DNA template molecules used in the methods provided herein include DNA molecules comprising, from 5' to 3', a first homology arm, a replacement DNA, and a second homology arm, wherein the homology arms containing sequences that are partially or completely homologous to genomic DNA (gDNA) sequences flanking a target site-specific endonuclease cleavage site in the gDNA. In certain embodiments, the replacement DNA can comprise an insertion, deletion, or substitution of 1 or more DNA base pairs relative to the target gDNA. In one embodiment, the donor DNA template molecule is double-stranded and perfectly base-paired through all or most of its length, with the possible exception of any unpaired nucleotides at either terminus or both termini. In another embodiment, the donor DNA template molecule is double-stranded and includes one or more non-terminal mismatches or non-terminal unpaired nucleotides within the otherwise double-stranded duplex. In an embodiment, the donor DNA template molecule that is integrated at the site of at least one double-strand break (DSB) includes between 2-20 nucleotides in one (if single-stranded) or in both strands (if double-stranded), e. g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, or 20 nucleotides on one or on both strands, each of which can be base-paired to a nucleotide on the opposite strand of the targeted integration site (in the case of a perfectly base-paired double-stranded polynucleotide molecule). Such donor DNA templates can be integrated in genomic DNA containing blunt and/or staggered double stranded DNA breaks by homology-di-

rected repair (HDR) or microhomology-mediated end joining (MMEJ). In certain embodiments, a donor DNA template homology arm can be about 20, 50, 100, 200, 400, or 600 to about 800, or 1000 base pairs in length. In certain embodiments, a donor DNA template molecule can be delivered to a plant cell in a circular (e.g., a plasmid or a viral vector including a geminivirus vector) or a linear DNA molecule. In certain embodiments, a circular or linear DNA molecule that is used can comprise a modified donor DNA template molecule comprising, from 5' to 3', a first copy of the target sequence-specific endonuclease cleavage site sequence, the first homology arm, the replacement DNA, the second homology arm, and a second copy of the target sequence-specific endonuclease cleavage site sequence. In other embodiments, DNA templates suitable for NHEJ insertion will lack homology arms that are partially or completely homologous to genomic DNA (gDNA) sequences flanking a target site-specific endonuclease cleavage site in the gDNA. Compositions comprising the donor templates can be delivered to the plant and/or meristem cells of the plant by particle mediated delivery, and any other direct method of delivery, such as but not limiting to, Agrobacterium-mediated transformation, polyethylene glycol (PEG)-mediated transfection to protoplasts, whiskers mediated transformation, electroporation, particle bombardment, and/or by use of cell-penetrating peptides. The donor DNA templates may be present transiently in the cell or it could be introduced via a viral replicon (e.g., a geminivirus replicon). Geminivirus DNA replicons suitable for delivery of donor DNA templates to plants include a Beet Yellow Dwarf Virus replicon (Baltes, N.J. et al. Plant Cell vol. 26,1(2014): 151-63. doi:10.1105/tpc.113.119792).

RNA guided nucleases can be provided to at least the meristem cell by a variety of methods that include stable expression with an integrated transgene, expression from a viral vector, or transient expression such as by introducing an RNA that encodes the RNA guided nuclease or an that RNA that encodes the RNA guided nuclease that is operably linked an MTS. In certain embodiments, an active form of the RNA guided nuclease is predominantly localized in meristem tissue of the plant. Delivery of RNAs encoding the RNA guided nucleases or DNAs then encode those RNAs to the plant and/or meristem cells of the plant can be achieved by particle mediated delivery, and any other direct method of delivery, such as but not limiting to, Agrobacterium-mediated transformation, polyethylene glycol (PEG)-mediated transfection to protoplasts, whiskers mediated transformation, electroporation, particle bombardment, and/or by use of cell-penetrating peptides. In certain embodiments, such predominant localization of the RNA guided nuclease can result in at least about 60%, 70%, 80%, 90%, 95%, 98%, or 99% of the active form of the RNA guided nuclease in the plant being localized in the meristem. In certain embodiments, the nucleic acid encoding the RNA guided nuclease can be delivered directly to the meristem by methods that include use of biolistic devices (e.g., as in U.S. 20200123554). In certain embodiments, the RNA guided nuclease can be operably linked to a vegetative stage, meristem-preferred or meristem-specific promoter including: (i) a pAt.Erecta, At.PNH, At.AN3, or At.MYB17 promoter or functional fragment thereof from Arabidopsis; (ii) a promoter or functional fragment thereof from a Glyma10g38730, Glyma09g27950, Glyma06g05900, or Glyma17g34380 soybean gene; or (iii) receptor like kinase (RLK) gene promoters from a PGSC0003DMP400032802 or PGS C0003DMP400054040 gene of potato. Such vegetative stage, meristem-preferred or meristem-specific pro-

moters are set forth in U.S. 20190300890, which is incorporated herein by reference in its entirety. In certain embodiments, expression of the RNA guided nuclease can be increased in floral meristems of maize plants by operable linkage to a floral meristem-enhanced promoters that include Zap1a, Zap1b, ZLF1, ZLF2, or ZMM4 endogenous genes (Dong et al. 2012 PLoS ONE 7(8):e43450). Alternatively, the RNA guided nuclease can be expressed in meristems and tissues other than the vascular tissues to mitigate cleavage of an RNA molecule comprising the gRNA and the MTS during transit from the site of contact to the meristem.

In some embodiments, a plant expressing transgenically a Cas polypeptide may be genome edited by delivery of a cargo containing only guide RNAs suitable for the transgenically expressed Cas polypeptide.

The RNA sequences are generally made and assembled at first in DNA form as RNA expressing vectors using recombinant DNA technology. RNA expression is done in vitro, and purified according to well established methods. Addition of RNA 5' caps and polyA tails to mRNAs can be performed according to methods established in the literature. Alternatively, some RNAs designed as described can be purchased from commercial providers.

A substantially purified RNA composition is understood to comprise a high concentration of an RNA molecule of interest, although in some cases it may comprise two distinct RNAs. For example, one RNA may comprise a Cas nuclease while another may comprise a corresponding guide or guide array. In addition, a substantially purified RNA composition may comprise other added components, such as a pH buffer, salt, surfactants, and/or RNase inhibitors.

Plants can be effectively contacted with the RNA vectors in many ways. Often it will be convenient to load them into the phloem of plants through the leaves, for example by nicking a leaf and submerging the injured tissue into a solution of substantially purified RNAs. Other avenues are also possible, such as by injection into the stems with a needle or use of a handheld biolistics device. In some embodiments, a surfactant is added to the purified RNA, and the liquid is applied to a tissue like embryonic shoot, leaf, stem, or inflorescence, with or without slight injury such as scratching.

The RNAs are often applied at the vegetative stage of the life cycle of a plant, so as to reach vegetative meristems before they convert to floral meristems. In some cases, however, it may be convenient to apply the vectors, RNA molecules, or compositions comprising the RNA molecules or vectors, to floral meristems, especially at early stages of differentiation. In certain embodiments, a soybean plant is contacted at the vegetative stage with a composition comprising the RNA molecules or vectors at vegetative stage Ve, V1, or V2 to about the V4 V(n) stage where 1, 2, 3, 4, or n is the number of trifoliate leaves (Soybean Growth and Development, M. Licht, 2014, Iowa State University Extension and Outreach, PM 1945). In certain embodiments, a maize plant is contacted at the vegetative stage with a composition comprising the RNA molecules or vectors at vegetative stage Ve, V1, or V2 to about the V4 V(n) stage (Corn Growth Stages, M. Licht, Iowa State University Extension and Outreach, on the <https://www.crops.extension.iastate.edu/encyclopedia/corn-growth-stages>”).

Very often, mutated seeds from plants edited with the reagents and methods described here are collected for phenotypic characterization. In some cases, pollen from edited plants is used in crosses with other individuals, or mutated individuals are pollinated with pollen of unedited plants or wildtype plants.

There are numerous plant-endogenous targets (i.e., DNA sequence targets) for genome editing. Any defective allele found in elite germplasm can get edited to a non-deleterious version. The methods presented here can be applied to a promoter bashing or fine-tuning approach, to create a range of phenotypes based on promoter alterations of a gene of a certain sequence or gene of interest (Rodriguez-Leal et al., Cell. 2017 Oct. 5; 171(2):470-480).

Editing of coding sequences can be made using the methods disclosed herein to increase the level of preselected amino acids in the encoded polypeptide. For example, the gene encoding the barley high lysine polypeptide (BHL) is derived from barley chymotrypsin inhibitor, U.S. application Ser. No. 08/740,682, filed Nov. 1, 1996, and WO 98/20133, the disclosures of which are herein incorporated by reference. Other proteins include methionine-rich plant proteins such as from sunflower seed (Lilley et al. (1989) Proceedings of the World Congress on Vegetable Protein Utilization in Human Foods and Animal Feedstuffs, ed. Applewhite (American Oil Chemists Society, Champaign, Ill.), pp. 497-502; herein incorporated by reference); corn (Pedersen et al. (1986) J. Biol. Chem. 261:6279; Kiriha et al. (1988) Gene 71:359; both of which are herein incorporated by reference); and rice (Musumura et al. (1989) Plant Mol. Biol. 12:123, herein incorporated by reference). Other agronomically important genes encode latex, Floury 2, growth factors, seed storage factors, and transcription factors.

The methods disclosed herein can be used to modify herbicide resistance traits including genes coding for resistance to herbicides that act to inhibit the action of acetolactate synthase (ALS), in particular the sulfonylurea-type herbicides (e.g., the acetolactate synthase (ALS) gene containing mutations leading to such resistance, in particular the S4 and/or Hra mutations), genes coding for resistance to herbicides that act to inhibit action of glutamine synthase, such as phosphinothricin or basta (e.g., the bar gene); glyphosate (e.g., the EPSPS gene and the GAT gene; see, for example, U.S. Publication No. 20040082770 and WO 03/092360); or other such genes known in the art. The bar gene encodes resistance to the herbicide basta, the nptII gene encodes resistance to the antibiotics kanamycin and genetecin, and the ALS-gene mutants encode resistance to the herbicide chlorsulfuron. Additional herbicide resistance traits are described for example in U.S. Patent Application 2016/0208243, herein incorporated by reference.

Sterility genes can also be modified and provide an alternative to physical detasseling. Examples of genes used in such ways include male tissue-preferred genes and genes with male sterility phenotypes such as QM, described in U.S. Pat. No. 5,583,210. Other genes include kinases and those encoding compounds toxic to either male or female gametophytic development. Additional sterility traits are described for example in U.S. Patent Application 2016/0208243, herein incorporated by reference.

Genome editing can also be used to make haploid inducer lines as disclosed in WO2018086623 and U.S. 20190292553.

The quality of grain can be altered by modifying genes encoding traits such as levels and types of oils, saturated and unsaturated, quality and quantity of essential amino acids, and levels of cellulose. In corn, modified hordothionin proteins are described in U.S. Pat. Nos. 5,703,049, 5,885,801, 5,885,802, and 5,990,389.

Commercial traits can also be altered by modifying a gene or that could increase for example, starch for ethanol production, or provide expression of proteins. Another impor-

tant commercial use of modified plants is the production of polymers and bioplastics such as described in U.S. Pat. No. 5,602,321. Genes such as .beta.-Ketothiolase, PHBase (polyhydroxybutyrate synthase), and acetoacetyl-CoA reductase (see Schubert et al. (1988) J. Bacteriol. 170:5837-5847) facilitate expression of polyhydroxyalkanoates (PHAs).

Exogenous products include plant enzymes and products as well as those from other sources including prokaryotes and other eukaryotes. Such products include enzymes, cofactors, hormones, and the like. The level of proteins, particularly modified proteins having improved amino acid distribution to improve the nutrient value of the plant, can be increased. This is achieved by the expression of such proteins having enhanced amino acid content.

The methods disclosed herein can also be used for modification of native plant gene expression to achieve desirable plant traits. Such traits include, for example, disease resistance, herbicide tolerance, drought tolerance, salt tolerance, insect resistance, resistance against parasitic weeds, improved plant nutritional value, improved forage digestibility, increased grain yield, cytoplasmic male sterility, altered fruit ripening, increased storage life of plants or plant parts, reduced allergen production, and increased or decreased lignin content. Genes capable of conferring these desirable traits are disclosed in U.S. Patent Application 2016/0208243, herein incorporated by reference.

The present disclosure may be used for genomic editing of any plant species, including, but not limited to, monocots and dicots (i.e., *monocotyledonous* and *dicotyledonous*, respectively). Examples of plant species of interest include, but are not limited to, corn (*Zea mays*), *Brassica* sp. (e.g., *B. napus*, *B. rapa*, *B. juncea*), particularly those *Brassica* species useful as sources of seed oil, alfalfa (*Medicago sativa*), rice (*Oryza sativa*), rye (*Secale cereale*), sorghum (*Sorghum bicolor*, *Sorghum vulgare*), camelina (*Camelina sativa*), millet (e.g., pearl millet (*Pennisetum glaucum*), proso millet (*Panicum miliaceum*), foxtail millet (*Setaria italica*), finger millet (*Eleusine coracana*)), sunflower (*Helianthus annuus*), quinoa (*Chenopodium quinoa*), chicory (*Cichorium intybus*), lettuce (*Lactuca sativa*), safflower (*Carthamus tinctorius*), wheat (*Triticum aestivum*), soybean (*Glycine max*), tobacco (*Nicotiana tabacum*), potato (*Solanum tuberosum*), peanuts (*Arachis hypogaea*), cotton (*Gossypium barbadense*, *Gossypium hirsutum*), sweet potato (*Ipomoea batatas*), cassava (*Manihot esculenta*), coffee (*Coffea* spp.), coconut (*Cocos nucifera*), pineapple (*Ananas comosus*), citrus trees (*Citrus* spp.), cocoa (*Theobroma cacao*), tea (*Camellia sinensis*), banana (*Musa* spp.), avocado (*Persea americana*), fig (*Ficus casica*), guava (*Psidium guajava*), mango (*Mangifera indica*), olive (*Olea europaea*), papaya (*Carica papaya*), cashew (*Anacardium occidentale*), macadamia (*Macadamia integrifolia*), almond (*Prunus amygdalus*), sugar beets (*Beta vulgaris*), sugarcane (*Saccharum* spp.), oil palm (*Elaeis guineensis*), poplar (*Populus* spp.), eucalyptus (*Eucalyptus* spp.), oats (*Avena sativa*), barley (*Hordeum vulgare*), vegetables, ornamentals, and conifers.

The embodiments described methods and reagents can have many advantages over other known solutions. The techniques presented generally bypass callus induction or tissue culture that are necessary for alternative or widely practiced genome editing procedures, thus speeding up (i.e., accelerating) and lowering or reducing the cost of the process of producing plants with targeted mutations. Epigenetic resetting (i.e., interference) is also eliminated. The

editing can be performed in individuals of an elite genetic background, making lengthy backcrossing schemes unnecessary.

Embodiments

Various embodiments of the compositions, vectors, recombinant DNAs, RNAs, and methods provided herein are set forth in the following set of numbered embodiments.

1. A composition comprising at least one RNA molecule comprising a cargo segment fused to a meristem transport segment (MTS), wherein the cargo segment comprises one or more guide RNAs for an RNA-guided nuclease or wherein the cargo segment comprises RNA encoding a TALEN or ZFN protein.

2. The composition according to embodiment 1, wherein the guide RNA is flanked by or comprises processing elements.

3. The composition according to embodiment 2, wherein the processing elements are direct repeat sequences of the bacterial CRISPR array of the RNA-guided nuclease or are direct repeat sequences that are processed by the RNA-guided nuclease.

4. The composition according to embodiment 3, wherein the cargo segment comprises a plurality of guide RNAs.

5. The composition according to embodiments 3 or 4, wherein the guide RNAs and the direct repeat sequences of the bacterial CRISPR array are for a Cas12a or a Cas12j RNA-guided nuclease.

6. The composition according to embodiment 1, wherein the composition comprises both a first and a second RNA molecule each comprising a cargo segment fused to an MTS, wherein the cargo segment of the first RNA molecule comprises guide RNAs which are distinct from the guide RNAs of the second RNA molecule, optionally wherein the guide RNAs of the first and second RNA molecule are flanked by or comprise processing elements which are processed by different RNA-guided nucleases.

7. The composition according to any one of embodiments 1 to 6, wherein the cargo segment does not contain an RNA-guided nuclease polypeptide-encoding sequence.

8. The composition according to any one of embodiments 1 to 6, wherein the cargo segment further comprises an RNA-guided nuclease polypeptide-encoding sequence.

9. The composition according to embodiment 8, wherein RNA-guided nuclease polypeptide-encoding sequence can be translated in a plant cell cytosol.

10. The composition according to embodiment 8 or 9, wherein the RNA molecule further comprises at least one polyA region, wherein the polyA region is 3' of the RNA-guided nuclease polypeptide-encoding sequence, and 5' of the guide RNA and/or wherein the polyA region is at the 3' end of the RNA molecule.

11. The composition according to any one of embodiments 1 to 10, wherein the composition comprises both a first and a second RNA molecule each comprising a cargo segment fused to an MTS, wherein at least the first RNA molecule comprises a cargo sequence further comprising an RNA-guided nuclease polypeptide-encoding sequence, wherein the cargo segment of the first RNA molecule comprises guide RNAs which are distinct from the guide RNAs of the second RNA molecule.

12. The composition according to embodiment 11, wherein the guide RNAs of the first and second RNA molecule are flanked by or comprise processing elements which are processed by different RNA-guided nucleases, and optionally wherein the processing elements in the first

RNA molecule are not recognized by the RNA-guided nuclease polypeptide encoded by the first RNA molecule.

13. The composition according to any one of embodiments 1 to 12, wherein the MTS comprises:

- (i) a Flowering Time (FT)-derived sequence, optionally wherein the FT-derived sequence is SEQ ID NO: 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, or 24, a meristem transport-competent (MTC) ortholog thereof, a MTC variant thereof, and/or a MTC fragment thereof;
- (ii) a tRNA like sequence (TLS), optionally wherein the TLS sequence comprises SEQ ID NO: 29 or 30, a meristem transport-competent (MTC) ortholog thereof, a MTC variant thereof, MTC fragment thereof, and/or an RNA hairpin comprising a first stem of 8 to 12 nucleotides, at least one variable bulge, a second stem of 4 to 7 nucleotides, and a variable loop;
- (iii) a GAI sequence, optionally wherein the GAI sequence comprises SEQ ID NO: 26, a meristem transport-competent (MTC) ortholog thereof, a MTC variant thereof, and/or a MTC fragment thereof;
- (iv) a BEL5 sequence optionally wherein the BEL5 sequence comprises SEQ ID NO: 28, a meristem transport-competent (MTC) ortholog thereof, a MTC variant thereof, and/or a MTC fragment thereof;
- (v) a CmNACP sequence optionally wherein the CmNACP sequence comprises SEQ ID NO: 25, a meristem transport-competent (MTC) ortholog thereof, a MTC variant thereof, and/or a MTC fragment thereof; or
- (vi) a LeT6 sequence optionally wherein the LeT6 sequence comprises SEQ ID NO: 27, a meristem transport-competent (MTC) ortholog thereof, a MTC variant thereof, a MTC fragment thereof.

14. The composition according to embodiment 13, wherein the MTS comprises a Flowering Time (FT)-derived sequence of SEQ ID NO: 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, or a meristem transport-competent (MTC) fragment thereof.

15. The composition according to any one of embodiments 1 to 14, wherein the MTS is located 3' of the cargo segment.

16. The composition according to any of embodiments 1 to 15, further comprising RNase inhibitors.

17. The composition according to any one of embodiments 1 to 16, wherein the RNA molecule is a substantially purified RNA molecule.

18. The composition according to any one of embodiments 1 to 17, wherein the RNA molecule is not operably linked to a viral vector RNA and/or associated with a viral protein.

19. A meristem-delivery vector comprising a cargo segment fused to a meristem transport segment (MTS), wherein the cargo segment comprises one or more guide RNAs for an RNA-guided nuclease.

20. A recombinant DNA having a sequence capable of producing as a transcript a vector according to embodiment 19, or producing an RNA that can be purified and combined with one additional component to form a composition according to any one of embodiments 1 to 18.

21. A method of producing a plant with an altered genome comprising

- (i) contacting a plant with at least a first composition according to any of embodiments 1 to 18, and
- (ii) retrieving a progeny of the plant, wherein the progeny has an altered genome.

22. The method according to embodiment 21, wherein contacting comprises phloem loading.

23. The method according to embodiment 21 or 22, wherein the contacting with the composition occurs at the vegetative stage of the plant life cycle.

24. The method according to any one of embodiments 21 to 23, wherein contacting comprises contacting the plant with the first composition, and after a time interval contacting the plant with a second composition according to any one of embodiments 1 to 18, wherein the guide RNAs in the cargo segment of the RNA molecule in the first composition are different than the guide RNAs in the second cargo segment of the RNA molecule in the second composition.

25. The method according to embodiment 24, wherein the time interval is about 18 or 20 to about 24, 30, or 36 hours.

26. The method according to any one of embodiments 21 to 25, wherein the guide RNA(s) of the RNA molecule are flanked by or comprise processing elements which are processed by the RNA-guided nuclease.

27. The method according to any one of embodiments 21 to 26, wherein:

- (i) wherein the RNA molecule does not contain an RNA-guided nuclease polypeptide-encoding sequence; and
- (ii) wherein the plant comprises a polynucleotide encoding the RNA-guided nuclease, optionally wherein the polynucleotide is integrated into the genome of the plant and/or optionally wherein an active form of the RNA guided nuclease is predominantly localized in meristem tissue of the plant.

28. The method of embodiment 27, wherein the RNA-guided nuclease is encoded by a DNA molecule, optionally wherein the DNA molecule is integrated into the genome of the plant, optionally wherein the DNA molecule is operably linked to a promoter which is preferentially expressed in target plant cells, and/or optionally wherein the target plant cells are meristem cells.

29. The method according to any one of embodiments 21 to 28, wherein the composition comprises both a first and a second RNA molecule each comprising a cargo segment fused to an MTS, wherein the cargo segment of the first RNA molecule comprises guide RNAs which are distinct from the guide RNAs of the second RNA molecule, optionally wherein the guide RNAs of the first and second RNA molecule are flanked by or comprise processing elements which are processed by different RNA-guided nucleases.

30. The method according to embodiment 29, wherein the composition comprises both a first and a second RNA molecule each comprising a cargo segment fused to an MTS, wherein at least the first RNA molecule comprises a cargo sequence further comprising an RNA-guided nuclease polypeptide-encoding sequence, wherein the cargo segment of the first RNA molecule comprises guide RNAs which are distinct from the guide RNAs of the second RNA molecule.

31. The method according to embodiment 29, wherein the guide RNAs of the first and second RNA molecule are flanked by or comprise processing elements which are processed by different RNA-guided nucleases, and optionally wherein the processing elements in the first RNA molecule are not recognized by the RNA-guided nuclease polypeptide encoded by the first RNA molecule.

32. A plant comprising:

- (i) an RNA molecule comprising a cargo segment fused to a meristem transport segment, wherein the cargo segment comprises one or more guide RNAs for an RNA-guided nuclease or a vector encoding the RNA molecule or wherein the cargo segment comprises RNA encoding a TALEN or ZFN protein; and,

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(ii) a DNA molecule or RNA molecule encoding the RNA-guided nuclease.

33. The plant according to embodiment 32, wherein the cargo segment does not contain a sequence encoding the RNA-guided nuclease.

34. The plant according to embodiment 32 or 33, wherein the cargo segment comprises a plurality of guide RNAs.

35. The plant according to any one of embodiments 32 to 34, wherein the guide RNAs and the direct repeat sequences of the bacterial CRISPR array are for a Cas12a or a Cas12j RNA-guided nuclease.

36. The plant according to any one of embodiments 32 to 35, wherein the plant comprises both a first and a second RNA molecule each comprising a cargo segment fused to an MTS, wherein the cargo segment of the first RNA molecule comprises guide RNAs which are distinct from the guide RNAs of the second RNA molecule, optionally wherein the guide RNAs of the first and second RNA molecule are flanked by or comprise processing elements which are processed by different RNA-guided nucleases.

37. The plant according to any one of embodiments 32, or 34 to 36, wherein the cargo segment contains a sequence encoding a Cas12a or a Cas12j RNA-guided nuclease, optionally wherein the Cas12a RNA-guided nuclease comprises SEQ ID NO: 53 or 55, or optionally wherein the Cas12j RNA-guided nuclease comprises SEQ ID NO: 57, 59, or 61.

38. The plant according to any one of embodiments 32, or 34 to 36, wherein the cargo segment further comprises an RNA-guided nuclease polypeptide-encoding sequence, optionally wherein a Cas12a or a Cas12j RNA-guided nuclease is encoded.

39. The plant according to embodiment 37 or 38, wherein RNA-guided nuclease polypeptide-encoding sequence can be translated in a plant cell cytosol.

40. The plant according to any one of embodiments 37, 38, or 39, the RNA molecule further comprising a polyA region, wherein the polyA region is 3' of the RNA-guided nuclease polypeptide-encoding sequence, and 5' of the guide RNA.

41. The plant according to any one of embodiments 32, or 34 to 40, wherein the composition comprises both a first and a second RNA molecule each comprising a cargo segment fused to an MTS, wherein at least the first RNA molecule comprises a cargo sequence further comprising an RNA-guided nuclease polypeptide-encoding sequence, wherein the cargo segment of the first RNA molecule comprises guide RNAs which are distinct from the guide RNAs of the second RNA molecule.

42. The plant according to embodiment 41, wherein the guide RNAs of the first and second RNA molecule are flanked by or comprise processing elements which are processed by different RNA-guided nucleases, and optionally wherein the processing elements in the first RNA molecule are not recognized by the RNA-guided nuclease polypeptide encoded by the first RNA molecule.

43. The plant according to any one of embodiments 32 to 42, wherein the MTS comprises:

(i) a Flowering Time (FT)-derived sequence, optionally wherein the FT-derived sequence is SEQ ID NO: 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, or a meristem transport-competent (MTC) fragment thereof;

(ii) a tRNA like sequence (TLS), optionally wherein the TLS sequence comprises SEQ ID NO: 29, SEQ ID NO: 30, a MTC fragment thereof, and/or an RNA hairpin

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comprising a first stem of 8 to 12 nucleotides, at least one variable bulge, a second stem of 4 to 7 nucleotides, and a variable loop;

(iii) a GAI sequence, optionally wherein the GAI sequence comprises SEQ ID NO: 26, a meristem transport-competent (MTC) ortholog thereof, a MTC variant thereof, and/or a MTC fragment thereof,

(iv) a BEL5 sequence optionally wherein the BEL5 sequence comprises SEQ ID NO: 28, a meristem transport-competent (MTC) ortholog thereof, a MTC variant thereof, and/or a MTC fragment thereof,

(v) a CmNACP sequence optionally wherein the CmNACP sequence comprises SEQ ID NO: 25, a meristem transport-competent (MTC) ortholog thereof, a MTC variant thereof, and/or a MTC fragment thereof; or

(vi) a LeT6 sequence optionally wherein the LeT6 sequence comprises SEQ ID NO: 27, a meristem transport-competent (MTC) ortholog thereof, a MTC variant thereof, a MTC fragment thereof.

44. The plant according to embodiment 43, wherein the MTS comprises a Flowering Time (FT)-derived sequence of SEQ ID NO: 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, or a meristem transport-competent (MTC) fragment thereof.

45. The plant according to any one of embodiments 32 to 44, wherein the MTS is located 3' of the cargo segment.

46. The plant according to any one of embodiments 32 to 45, wherein an active form of the RNA guided nuclease is predominantly localized in meristem tissue of the plant.

47. The plant of any one of embodiments 32, or 34 to 46, wherein the RNA-guided nuclease is encoded by a DNA molecule and optionally wherein the DNA molecule is integrated into the genome of the plant.

48. The plant of embodiment 47, wherein the DNA molecule encoding the RNA-guided nuclease is operably linked to a promoter which is preferentially expressed in target plant cells and optionally wherein the target plant cells are meristem cells.

49. A plant comprising an altered genome made by the method of any one of embodiments 21 to 31.

50. The use of the composition of any one of embodiments 1 to 18 to obtain a plant with an altered genome.

EXAMPLES

Example 1—RNA Design

The basic plasmid design to produce the editing message starts with a standard high copy plasmid that contains a multiple cloning sites downstream of the T7 promoter, such as pBluescript™ or pSP73. Each component can be easily introduced using an efficient assembly approach. The design consists of a plant codon optimized Cas12a coding sequence followed by the DR sequence of the Cas12a CRISPR array, in which the DNA-targeting spacer sequences are replaced a guide with soybean phytoene desaturase (PDS) gene as a visual marker (Du et al. J. Biotech 2016, 217:90-97; doi.org/10.1016/j.jbiotec.2015.11.005). The guide RNA region is followed by the an FT sequence derived from Arabidopsis (SEQ ID NO: 1). The DNA vector sequence ends in a unique restriction site to linearize the plasmid for runoff transcription. This arrangement enables production of high quantity editing mRNA.

Example 2—Production of the RNA Composition

To produce the mRNA for plant delivery the production vector above is linearized as template for in vitro transcrip-

tion to produce tens of micrograms of editing mRNA using a system such as mScript™ (CAMBIO, Cambridge, UK; on the world wide web <https://internet.site/cambio.co.uk/20/431/21/products/t7-mscript-standard-mrna-production-system/>). The product is cleaned up and characterized to make sure it is the expected size and to determine how much mRNA was produced. The purification process includes a DNAase treatment followed by a phenol chloroform extraction then ethanol precipitation and resuspension in RNase free water. RNAase inhibitor is also added (New England Biolabs, Ipswich, MA, USA; on the world wide web <https://internet.site/neb.com/products/m0314-rnase-inhibitor-murine#Product%20Information>) to stabilize the editing mRNA during uptake by the plant.

Example 3—Phloem Loading

The in vitro transcription reaction of Example 2 produces 50 micrograms of editing mRNA. It is suspended in a mix at 0.2 micrograms per microliter (10 micrograms mRNA in 50 microliters of RNase free water) in nuclease-free Eppendorf™ tubes (1.5 mL). These steps produce sufficient material for five replicates. A negative control contains everything but the editing mRNA. The soy plants are at the 2-3 trifoliate stage in small pots. Using sharp, clean & heat sterilized scissors to remove a leaf tip in the second trifoliate of each plant then the leaf tip is cut when submerged in sterile nuclease free water. Very gently the leaf is placed in the RNA solution and the setup stabilized so the plant can absorb the mRNA solution with no undue stress. Uptake of the editing mRNA takes several hours.

Example 4—Phenotyping

The treated leaves are removed from the editing mRNA tubes when the solution is depleted to minimize wounding. In 1-2 weeks for the intended phenotype will appear in new growth. The soy PDS knockout is lethal so the plants will likely not set seed, but the same method can be adapted to make non-lethal mutations that are transmissible through in the germline.

All cited patents and patent publications referred to in this application are incorporated herein by reference in their entirety. All of the materials and methods disclosed and claimed herein can be made and used without undue experimentation as instructed by the above disclosure and illustrated by the examples. Although the materials and methods of this disclosure have been described in terms of embodiments and illustrative examples, it will be apparent to those of skill in the art that substitutions and variations can be applied to the materials and methods described herein without departing from the concept, spirit, and scope of the invention. For instance, while the particular examples provided illustrate the methods and embodiments described herein using a specific plant, the principles in these examples are applicable to any plant of interest. All such similar substitutes and modifications apparent to those skilled in the art are deemed to be within the spirit, scope, and concept of the invention as encompassed by the embodiments of the inventions recited herein and the specification and appended claims.

SEQUENCE LISTING

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<213> ORGANISM: Arabidopsis thaliana

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<212> TYPE: DNA
<213> ORGANISM: Arabidopsis thaliana

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caccaacat tccactaaga ataacgtaca acaataggct acttctgcc aagtctgagc	180
taaagccatc cgcggttgta agtaaacac gagtcgatat cgggtggcagt gacatgaggg	240
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gggagtactt gcactggatg gtgacagata ttccagaaac aactagtgc aactttggcc	360
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gtcaaaggga aggtggatcc ggcggaagaa ggtttaggga agagtagaaa ccataggcca	600
ctgcatggtc aactataga aatatcatca ataattgtca ctatattgaa tcaatgcacc	660
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gagcttagtc tcttagcaga atatagcaca atattactag ta	762

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gtgtttgtgt tatttcgtca actgggtagg gagaccgtgt atgcaccagg atggcgccag	420
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<213> ORGANISM: Arabidopsis thaliana

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<212> TYPE: RNA

<213> ORGANISM: Arabidopsis thaliana

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uuggagacgu ucuugaucg uuuaauagau caaucacucu aaagguuacu uauggccaaa	240
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agauuggugg agaagaccuc aggaacuucu auacuuuggu uauuggugau ccagauguuc	360
caaguccuag caaccucac cuccgagaau aucuccauug guuggugacu gauauccug	420
cuacaacugg aacaaccuuu ggcaaugaga uugugugua cgaaaaucca agucccacug	480
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uuuagauggc uucuuccuuu auaaccaauu gauauugcau acucugauga gauuuauca	720
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<212> TYPE: RNA

<213> ORGANISM: Zea mays

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caccaaccau uccacuaaga auaacguaca acaauaggcu acuuucugcca agugcugagc	180
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cuuucuaac ccugguacug auugaccccg augcccaag uccaagccau ccaucacuaa	300
gggaguacuu gcacuggaug gugacagaua uuccagaaac aacuauguc aacuuuggcc	360
aagagcuauu auuuuauag aggccggacc caagaucugg cauccacagg cuggauuuug	420
ugcuguuccg ucaacuuggc agggggacag uuuuugcacc agaaugcgc cacaacuua	480
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cugcaugguc acacuaauaga aauaucauca auaaugugca cuauauugaa ucaaugcacc	660
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auuuucaaca cuaaagaauu ugcugaacuu uacaaccuug gauugccagu ugcugcuguc	480
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ccaucucgau uaggguguuu uacgacucga gggaauguua caaugggugu gagcucaaac	180
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ucacuuuggu uaugguggau cccgacgcuc cuagcccuag cgaucaccau cuaagagaau	300
acuugcauug guuagugacc gauauuccag cuacaaccga ggcaaccuuu ggacaagaga	360
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auauugcucu uauacucguu ucguaugaau cacugucuuu ucuucuuugu uuucuuuguu	780
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ccuacaaua uagagaagua gcaauaggcu gugaguucag acccucucag cuagucagcc	180
aaccuagggg ggacauugga ggggaugacu ugaggaccuu cuauacuug guuauaggug	240
accucagcgc uccaagcccc aguaauccga accuaaggga guacuuaau ugguugguga	300
cugauauucc agcaacuacu ggggcaacu ucggccaaga gauugugugu uaugagagcc	360
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 <213> ORGANISM: Malus domestica

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gggcaagaga ucguguguua ugaaagucca cggccaacag uggggauuca ucgcuuuguu	360
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uucaacacaa gagauuuugc agaacuuaa aaucuugguu uaccguugc ugcugucua    480
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<212> TYPE: RNA

<213> ORGANISM: Populus tremula

<400> SEQUENCE: 15

```

augucaaggg acagagaucc ucugagcggu ggcguguuu uaggggacgu gcuggacccc    60
uucacaaagu cuaucccgcu caggguacac uacaacucca gagagguaa caaugguugc    120
gagcucaaac ccucucaggu ugccaaccag ccgaggguug auauuggcgg ggaagaucua    180
aggaccuucu acacucuggu uauuggugac ccugaugcac ccagcccaag ugacccacg    240
cucagagaau auuugcauug guuggugacu gauauuccag caacaacggg ggcaagcuu    300
ggccaugaaa cugugucua ugagagcccg aggccgacga uggggauuca ucgguuuguu    360
uucgucuugu uccggcaacu gggcaggcaa acuguguauu cccugggug gcgccagaac    420

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 uucaacacca gagacuugc ugaggucuaa aaucuggau cgccgguggc ugcuguuuau 480

uucaacugcc agagggagag uggcucuggu gguaggaggc gauaa 525

<210> SEQ ID NO 16

<211> LENGTH: 534

<212> TYPE: RNA

<213> ORGANISM: Hordeum vulgare

<400> SEQUENCE: 16

auggccggga gggacaggga uccgcugguu gucggcaggg uuguggggga cgucugggac 60

cccuucgucc gaaccaccaa ccucaggguu accuucggga acagggccgu guccaacggc 120

ugcgagcuca agccguccau ggucgcccag cagccgaggg uggagguggg cggaauag 180

augaggaccu ucuacacguu cgugauggua gaccagaug cuccaagucc uagcgacccc 240

aaccuuagag aguaucucca cugguuggu acagauuucc cggguacaac uggggcgucg 300

uucgggcagg aggugaugug cuacgagagc ccucguccaa ccauggggau ccaccguuc 360

gugcugugc ucuuccagca gcuggggcgg cagacggugu acgccccggg guggcgccag 420

aacuucaaca ccagggacuu ugccgagcuc uacaaccucg gccagcccg ugcgcgcguc 480

uacuucacu gccagcgca ggccggcucc ggccggcagg ggauguacaa uuga 534

<210> SEQ ID NO 17

<211> LENGTH: 537

<212> TYPE: RNA

<213> ORGANISM: Hordeum vulgare

<400> SEQUENCE: 17

auggugggga gcagcaugca gcgcggggac ccgcuggugg uggggcgggg gaucggcgac 60

gugguggacc cguucgugc gcgggugggc cugcgggucg gcucagcguc cagggacgug 120

gccaacggcu gcgagcucc gccgucggc aucgcccacc agccgcgcgu cgaggucggc 180

ggcccgga ugcgcaccuu cuacaccug gugauggugg auccggaugc uccaagcccc 240

agcgacccca gccuaggga guacuugcac uggcugguca ccgacauccc ggccacgaca 300

ggagugucuu uugguaccga gguugugugc uacgagggcc cgcggccggg gcucgggauc 360

caccgacugg uguuccugcu cuuccagcaa cucggccgac agacggugua cgccccgggg 420

uggcggcaga acucagcac ccgcgacuuu gccgagcucu acaaccucgg ccugcccguc 480

gccgcgcguc acuucaacug ccagaggag accggaaccg gcgggagaag gauguga 537

<210> SEQ ID NO 18

<211> LENGTH: 847

<212> TYPE: RNA

<213> ORGANISM: Oryza sativa

<400> SEQUENCE: 18

ugcaccacac acaguacagc uagcagauga ccuagcuaga uagcugccuc uaucacagua 60

uauuugcucc cugcaacuug cugcugcugc aaugcuagc agcugcagcu aguaagcaaa 120

acuaaaacc uucaggguuu uuugcaagau cgauggccgg aaguggcagg gacagggacc 180

cucuuguggu ugguaggguu guggugaug ugcuggacgc guucgucgg agcaccaacc 240

ucaaggucac cuauggcucc aagaccgugu ccaauggcug cgagcucaag ccguccaugg 300

ucaccacca gccuagguc gaggucggcg gcaaugacau gaggacauuc uacacccuug 360

ugaugguaga cccagaugca ccaagcccaa gugacccuaa ccuaggagg uaucuacauu 420

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gguuggucac ugauauuccu gguacuacug cagcgucuuu ugggcaagag gugaugugcu	480
acgagagccc aaggccaacc auggggaucc accggcuggu guucgugcug uuccagcagc	540
uggggcgua gacaguguac gcgcccggu ggcgucagaa cuucaacacc aaggacuucg	600
ccgagcucua caaccucggc ucgccggugc ccgccgucua cuucaacugc cagcgcgagg	660
caggcuccgg cggcaggagg gucuacccu agcuaacgau gaucgccauc gaucugcugc	720
augcucacua ucaucaucca gcaugcuaua cauugcaggu ucagacaaau gaaaugauuc	780
ucgacacaca acauauauau gaugguguaa uuaauuugc aaauaaauag cugagcaagg	840
cuaaggu	847

<210> SEQ ID NO 19

<211> LENGTH: 866

<212> TYPE: RNA

<213> ORGANISM: Oryza sativa

<400> SEQUENCE: 19

ccugucacug uuuggcuagc uuaaccuucc ugacaucau ccucuggauu gaacggcagg	60
agauaccuaa gcuagcuagc aaucucuauc gaucuguuug uuuaucguu caguuaaagg	120
uuacugagaa augccuagag uuuuuccggc uagcuucau aguuaugggg uuagcugacc	180
uagauucaa gucuaauccu uuuaauuuu ugauuuaga uauccuaacg uuuuuaguua	240
gagguuuua auuugacaug gccggcagcg gcagggacga uccucuugug guuggcagga	300
uuguggguga ugugcuggau ccauucgucc ggaucacuaa ccucaguguc agcuauaggug	360
caaggauccu cuccaauagg ucgcagcuca agccguccau ggugacccaa cagcccaggg	420
ucguggucgg uggcaaugac augaggacgu ucuacacacu cgugauggua gaccgggag	480
cuccgagccc aagcaaccu aaccuuaggg aguaucuaa cuggcuggucc accgauauuc	540
cugguaccac uggagcaaca uuugggcaag agguagugug cuacgagagc ccaaggccaa	600
ccauggggau ccaccggcug guuucgugc uguuccagca gcuggggcgu cagacggugu	660
acgcaccggg gugggcccag aacuucagca ccagggaacu gcgcgagcuc uacaaccucg	720
gcucgccggg cgccaccguc uacuucacu gccagcgcg ggcgggcucc ggccggcagga	780
gggcuaccc cuagcuagcu acgcaugcca cccggccucc augcaugcag cagcuauagc	840
uaagcugaga ccugccuagc uguaua	866

<210> SEQ ID NO 20

<211> LENGTH: 848

<212> TYPE: RNA

<213> ORGANISM: Ipomoea nil

<400> SEQUENCE: 20

cacacacaca cacauauaua uacagagaaa gguuaguuu gaucgaggag cugagcuagc	60
uaggauccga aggggaacag uagaccuuu gguguugggg cgugugaucg gagacguugu	120
ggauccauuc acgagguccg uuagcuuag gguguuuac aaauacgagg uggauaucag	180
gaaugggugu gagaugaggc cuucucagcu caucaaccca ccuagggguu aaauccggcg	240
acacgaucuc cguacuuuc acacucuggu uaugguggau ccugaugcuc caaguccaac	300
cucuccaacc cugagggaau accuccacug guuggucacu gauauaccag gaacuacagg	360
agcaagcuuc ggcaaugaag cgauuuucua cgagccucca aggcgcuaa ugggaaucca	420
ccguuuugug uuugucuuu uccggcaacu uggccggcag acaguuuag caccgguuug	480
gcgccagaa uucaacacuc gaaacuugc ugagauuuc aaucuuuguu ugccaguggc	540

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cgucacuuac uuuaacggcc aaagggaggg uggcaccggc ggucgaucuc cggcagagcc	600
cugggcagcc gauuaauuac ccugcuccuu cccguuaauu ucaugcaugc augcaugcua	660
ucuauagcau aacauacaua uaguauauau cauaaaauaa uaagaccaca ugcauuuaca	720
uguuuauuuu ucccaugaau auauguuaaa guuguucuaug aagaacuacg uacuccauua	780
uauuacccuu uauauauggc aaugaagaug guuucaucuc uauuuagaag cuaaaaaaaa	840
aaaaaaaa	848

<210> SEQ ID NO 21
 <211> LENGTH: 798
 <212> TYPE: RNA
 <213> ORGANISM: Arabidopsis thaliana

<400> SEQUENCE: 21

uuuuuugaga uacuugagau ccaagauaaa uaugucuuua gucguagaga uccucuugug	60
gucggcagug uuguuggaga uguuucugau ccuuucacga gguuggucuc ucuuaagguc	120
acuuauaggcc auagagaggu uacuaauggc uuggaucuaa ggccuucua aguucugaac	180
aaaccaauag uggagauugg aggagacgac uucagaaauu ucuacaccuu gguuauaggug	240
gauccagaug ugccgagucc aagcaaccuu caccaacgag aaauaucucca cugguuggug	300
acugauauac cugccaccac uggaaaugcc uuuggcaaug agguuggugug cuacgagagu	360
ccacgucccc ccucgggaau ucaucguauu guguugguau uguuccggca acucggaaga	420
caaacgguuu augcaccggg guggcgccaa caguucaaca cucgugaguu ugcugagauc	480
uacaauucug gucuuccugu ggcugccucu uacuucaacu gccagaggga gaauggcugu	540
gggggaagaa gaacguagau gcguaccuac uuacguuac uaaauaucua aucguauau	600
auucccuuaa ugaaguauu aagcaucuau gucaauguaa uaagaauuaa aagauacgag	660
cuaaaaaaaa ugaugcauau gcugacaucg auguaaagua guuuacacuu uuaauguaau	720
aacuagguuu uaaccgcgg uacaccgca gacuauuuug uuuuuuaag auaaaaaua	780
uaauuuguuu agucgauu	798

<210> SEQ ID NO 22
 <211> LENGTH: 519
 <212> TYPE: RNA
 <213> ORGANISM: Glycine max

<400> SEQUENCE: 22

auggcacggg agaaccucu uguuuuuggu ggugugauug gggauuguu caaccuuuu	60
acaagcuccg uuucuuugac uguuucauac aaauuuaggc cgauuagcaa uggcuuggaa	120
cucaggcccu cucaaguugu uaaucgccu aggguuacug uuggugguga agaccuaagg	180
accuucuaa cucugguuau gguggaugca gaugcaccua gccuagcaa ccugucuuug	240
agggaaauacc uucacuggau ggugacagau auuccagcua ccacaaauagc aagcuuuggg	300
agagagguug uguuuuaua gagcccgaa ccuucaguag ggaaucaucg aaucguguuc	360
guuuuguuuc agcaauuggg cagagacacu gucaucaccc cagaauaggc ccauaauuuc	420
aauccagaa acuuugcuga aaauaauaac cuugcaccug uugcagcagc uuaugccaac	480
ugccaaagag agcgugguug cgguggaagg agauuuuaa	519

<210> SEQ ID NO 23
 <211> LENGTH: 901
 <212> TYPE: RNA

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<213> ORGANISM: *Zea mays*

<400> SEQUENCE: 23

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agagcacauc cguagugugu gcaugcauca cagucacaca cacacagcag aagaagaaga      60
aaccgaacga ggguuuagcu agcaaaaaua acagaagcaa gcaagcuagc uagagcuaag     120
gaucgagauc gagaucgacc gaccgacgac gaucagcuag cauggcgcg c uucguggauc     180
cgcuagguggu gggcgcggu g aucggcgagg ugguggaccu guucgugccu uccaucucca     240
ugaccgucgc cuaugauggc cccaaggaca ucagcaacgg cugccuccuc aagccguccg     300
ccaccgccgc gcccgcguc guccgcaucu ccggcgccgc caacgaccuc uacacgcuga     360
ucaugacgga ccccgau ggc ccuagcccc gcaaccgcac caugaggag uaccuccacu     420
ggauagugau uaacauacca ggaggaacag augcuacuaa aggugaggag gugguggagu     480
acaugggccc gcggcgccgc guggguaucc accgcuacgu gcuggugcug uucgagcaga     540
agacgcgcgu gcacgcggag gcccccgcg accgcgccaa cuucaagacg cgcgcguucg     600
cgggcgcgca cgagcucggc cucccccacug ccgucgucua cuucaacgcg cagaaggagc     660
ccgccagccg ccgcccguag cuagcagcuc cucucugagg caugccagau gcaugcgugu     720
gcgugcaggu gcaaccaccg cacugccggc ggcuauguau gaccggugaa uaaaaaguuu     780
uacugcaccg uaagcaugcu cgcgcguug cuauugguau auguuagcag uguggcaguc     840
uguauaguau agcauuucgc uugcaucuau gcacucuauu uuaguaugcg uacguguggu     900
u                                                                           901

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<210> SEQ ID NO 24

<211> LENGTH: 1069

<212> TYPE: RNA

<213> ORGANISM: *Zea mays*

<400> SEQUENCE: 24

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uggcaaaaac ccagcgcuuu gugccgccgc cguccgccgc cccucugcc cuuguacgcg      60
caccuagaca caucgucauc gaucaucaca cgcaaucgac acaagaaguu aaauaacagc     120
ccaaggacgc agagaucagc ugaucgagaa ggacuuguac uacuacucag uauugucguc     180
acaugcaca uauauguacau aaagagcuag cuaccugagc ucucccag gucgcguuga     240
ucgaucgauc auggcgcggu ucguggaccc gcugguggu gggcggguga ucggcgaggu     300
gguggaccug uucgugcccu ccgucuccau gaccgucgcc uauggcccca aagacauacg     360
caacggcugc cuccuacagc cguccgccac cgcgcgcgcg ccgucgcucc gcaucuccgg     420
ccgccgcgac gaccucuaca cgcugaucau gacggaccca gaugcgccua gccccagcga     480
cccgaccaug agggaguacc uccacuggau agugacuaac auaccaggag gaacggaugc     540
aaacaaaggu gaggaggu gggaguuacau gggcccgcg ccgcccgucc gaauccaccg     600
cuacgugcug gugcuguuuc agcagaagac gcgugugcac gcggagggu cgggugagcg     660
cgccaacuuc aacacacgcg cguucgcggc ggcgcacgag cucggccucc ccaccgccgu     720
cguguacuuc aacgcgcaga aagagccggc caaccaccgc cgcgcguagc uaguagcucc     780
aacaaggcg cgccagcuga gcugcgugcg ugcaaccac cacacagccg ccggcgaggg     840
cugccuauau gaccggcgaa uaaaaagcu uacugcaccg uccguaggcg uacucucugu     900
ugguauaugc uugucuucag gcucuugagu cuaucuacuu aaauuggguu accacugagu     960
aaugaagca guuggcgcuu cgaucauca uucuaauauc cguacguguc aaucuuuccu    1020
guuuccauca ucuugcauuu gaagacgc au ugguuuaca ccaaggugu                1069

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<210> SEQ ID NO 25
 <211> LENGTH: 1288
 <212> TYPE: RNA
 <213> ORGANISM: Cucurbita maxima

<400> SEQUENCE: 25

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gacuuuuuau ucaacaauuc cucucucucu cucucaacuu ccgaucaagu cucuccgccg      60
ucuuuucacc ggagcugaca auuccgauca uuuuuugcuu cccuuuuuuu uccggcaugc      120
aggaaccacc gccaaacgcc uuggauuugc ccccuggcuu cagaauccac cccaccgacg      180
aggagaucgu cacuuuuuac cugauacaua agaucaccga cgccgccuuc acugccaccg      240
ccaucggaga agcugaccug aauaagugug aaccuuggga uuugccacau aaagcuaaga      300
uggggggaaaa agaauuguau uuuuuuugcc agagagaccg gaaauauccg accgggauga      360
gaacgaaccg ggcgacucag accgguuacu ggaagcgac cgggaaagac aaggagauuc      420
ucaaggggaag aacgguucug gcugguauga agaaaacgcu gguuuuuuac aaaggaagag      480
cucccaaaag ugaagagacc aauggguuca ugcaugaauu ucgacucgaa cccaaauucu      540
uucaguuuuc ugguuuuucc aagccauua aggcugauug gguuguaugu cggguuuuuc      600
acaagaacac aacgaacacg gucggaguag ugaaaaagau ucaaacuucu gauuuuuucu      660
cuucucucucc accucuaaua gauccacaa cugcucauuc uccaaucagu ggcagauucg      720
auaaugguga agucaacugg agguuauacg uaccauucga uaaauaugca aaugauuacc      780
auuaucaucg gccuuuuuca ggcagcaaua cugcagugac aaugauuucg ucguaccgau      840
cgucugucucc cgacgacgaa uucucucacu ugaucacuu agacgucggu ggaacaaugu      900
caauggcggc ggcgacgaca acaacaacaa caacuaugga gugcaaaaua gaacaaguuu      960
cauggucaac gaugagcggu gugacaccgg agauauauc gucgauugac aacgaggcag      1020
cucucgaguu cugggacuac ugaaaaauga aaguagaugu uaugaucgaa caauggcgau      1080
gcuuuguuuu aaugggcau uucccauuu gaacguuuua acaaugauua auugauugcu      1140
aaauuuuuuu auuuuuuuuu uuugguuaca uaguccuuuu uggaaggaa uauuagaacu      1200
uucauggguu ugguuuuguu auuguauuga uauguagcaa ugugacauug uauauagcuu      1260
cuuauucuuu uauuuuaacc guugcaaa      1288

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<210> SEQ ID NO 26
 <211> LENGTH: 1964
 <212> TYPE: RNA
 <213> ORGANISM: Arabidopsis thaliana

<400> SEQUENCE: 26

```

uaauaaucuu uuuuuuucuu auuaccuucc ucucuuuuu uacaaauuuu uuuguuuuuu      60
gaagugguag uggagugaaa aaacaaaucc uaagcagucc uaaccgaucc ccgaagcuua      120
agauucuuca ccuucccaaa uaaagcaaaa ccuagauccg acauugaagg aaaaaccuuu      180
uagauccauc ucugaaaaaa aaccaaccuu gaagagagau caucauauc aucaucaaga      240
uaagaagacu augaugauga augaagaaga cgacgguaac ggcauggaug agcuucuaagc      300
uguucuuugu uacaagguaa ggucaucgga aauggcugau guugcucaga aacucgagca      360
gcuugaaguu augaugucua auguuaaaga agacgaucuu ucuaaacucg cuacugagac      420
uguucacuuu aaucggcgcg agcuuuacac guggcuugau ucuauugcuu ccgaccuuua      480
uccuccgucg ucuaacgcgg aguacgaucu uaaagcuuu cccggugacg cgaauucuaa      540

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ucaguucgcu aucgauucgg cuucucuguc uaaccaaggc ggcggaggag auacguauac	600
uacaaacaag cgguugaaau gcucaaacgg cgucguggaa accaccacag cgacggcuga	660
gucaacucgg cauguugucc ugguugacuc gcaggagaac ggugugcguc ucguucacgc	720
gcuuuuggcu ugcgcugaag cuguucagaa ggagaauucg acugugcgcg aagcucuggu	780
gaagcaaac ggauucuuag cuguuucuca aaucggagcu augagaaaag ucgcuaacuua	840
cuucgccgaa gcucucgcgc ggcggauuaa ccgucucucu ccgucgcaga guccaaucga	900
ccacucucuc uccgauacuc uucagaugca cuucucagag acuuguccuu aucucaaguu	960
cgucacucuc acggcgaauc aagcgauucu cgaagcuuuu caagggaaga aaagaguuca	1020
ugucauugau uucucuauga gucaaggucu ucaauggccg gcgcuaugc aggcucuugc	1080
gcuucgaccu ggugguccuc cuguuuuccg guuaaccgga auugguccac cggcaccgga	1140
uaauuucgaa uaucuucaug aaguugggug uaagcuggcu cauuuagcug aggcgauuca	1200
cgugaguuu gaguacagag gauuuguggc uaacacuuua gcugaucuuug augcuucgau	1260
gcuugagcuu agaccaagug agauugaau uguugcgguu aacucuguuu ucgagcuuca	1320
caagcucuu ggacgaccug gugcgauca uaagguucuu ggugugguga aucagauuaa	1380
accggagauu uucacugugg uugagcagga aucgaaccau aauguccga uuucuuaga	1440
ucgguuuacu gagucguugc auuauuacuc gacguuguuu gacucguugg aagguguacc	1500
gaguggucua gacaaggua ugucggaggu uuacuugggu aaacagaucg gcaacguugu	1560
ggcuugugau ggaccugacc gaguugagcg ucaugaaacg uugagucagu ggaggaccg	1620
guucgggucu gcuggguuug cggcugcaca uauugguucg aaugcguuu agcaagcgag	1680
uauuguuuug gcucuguuca acggcgguu ggguaucgg guggaggaga gugacggcug	1740
ucucauguug gguuggcaca cagcaccgcu cauagccacc ucggcuugga aacucuccac	1800
cauuuagaug guggcucaau gaauugaucg guugaaccgg uuaugaugau agauuuccga	1860
ccgaagccaa acuaaaucuu acuguuuuuc ccuuugucac uuguuaagau cuuauuuuuc	1920
auuauuuuag guaaauagaa auuucuaaaa uuacucacac uggc	1964

<210> SEQ ID NO 27

<211> LENGTH: 1556

<212> TYPE: RNA

<213> ORGANISM: Lycopersicon esculentum

<400> SEQUENCE: 27

aaagaaaaa ggaauuuugu guuuuugcuu uuuuuucuga cuaguaguau ugcuaacuau	60
guauuccauu aaggauuugc ugugaaaaag ccugauauca guaagcauaa aacucgggag	120
aucacuuaca cacacacaca ccuccuuaa aaagagaaga gagauuuacu guuaaacaga	180
gguuuuuuuc cauucuuuu uuuuuuucag ugugugugug agagaaagag augauuuuca	240
uaggcacaaa caauuagaaa ggaacaaaau uuagagugaa gaagaaagug ugugagagaa	300
uaauggaggg ugguucuaug ggaauacua guacauuug uuuuauaug augggauaug	360
gagauauga aaacaacaac aacaacaau gaaugguua uggaauugga aauggaaug	420
uaacaauuug ugcuccucca augaugauga ugaugccucc uccuccuccu ucuuuaacua	480
acaauaaca ugcagaaaca agcaacaaca acauccuuu ucuuccuuuc auggacaaca	540
acaacaaca uauuccucaa gaagacaaca acucuuuuc ucuuccauc aagucaaaga	600
uuauggcuca uccucacuac caucgucucu ugacugcuua ucucauuugu caaaagauag	660
gagcuccgcc agaaguggug gcaaggcuag aggaaauaug ugccacguca gcaacaauug	720

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gccguagcag uaguaguagu ggugguggaa ucauuggaga agauccugca cuagaucagu	780
ucauggaggc uuauugugag augcugacaa aaauugaaca agaaccuca aaaccuua	840
aggaagccau gguuuuuuu ucaagaauug agugucaguu caaagcuua acucugcac	900
cuaauucuc ucaugaauu gcuuugggagc aggcaaugga uagaaugga ucaucugaug	960
aagagguuga cgugaauaac aguucaucg acccccaggc ugaggauaga gagcucaaag	1020
gucaauuguu gcguaaguac agcgguuacu uggaagccu uaagcaggag uucaugaaga	1080
agaggaagaa aggcaagcug ccuaaggag caaggcaaca auugguggau ugguugcuua	1140
gacauuuua auggccauu ccaucggau cucagaagcu ugcacugcu gaaucaacgg	1200
gauuggacca gaagcaaua acaacuggu uuaucaaua aagaagagg cauugaaac	1260
caucagaaga uaugcaguuu guugugaugg augcugcuu uccacauuac uauauggaua	1320
auguucugc uaaccuuuc ccaauggau ugacaccuc ucuccugc auuaagauuu	1380
gucauuuaa auaucaagga uguuuuuuu auuugcauu uacuugugug cauguaguag	1440
uacaagcuau ugugacacaa ucaacuuuu auuagaccaa auauuaaag ugcuuuauu	1500
agaucuuuc auuaucauc uuaauuauug aauuaauag uuuguacuug cuaaaa	1556

<210> SEQ ID NO 28

<211> LENGTH: 2735

<212> TYPE: RNA

<213> ORGANISM: Solanum tuberosum

<400> SEQUENCE: 28

caugcagaga uaaaaauua gaucagucug acaagaaggc aacuucuaa agcuuagaga	60
gcuaccacc gaagauagac aguuguuac auguacuguu auagauaaa ggagaaaucc	120
gaagaagaaa gaauuuuuu ugcagauaug uacuaucaag gaaccucgga uauuacuaa	180
auacaagcug aucaucaaca acgucauuu caugggaaua guauuaaua uauuuuacg	240
acacuuuuu ugaugaacc uaacaauuau augcaaggcu acacuucuc ugacacacag	300
cagcagcagc aguucuuuu ccugaauuc ucaccagcag caagcaacgc gcuuugccau	360
gcgaauuac aacacgcgc gcugcaacag cagcacuuu ucggugugcc ucuccggca	420
guaguuuug acgaucagau cauaucau ggacuuuac agcgcaugug gaacaaccaa	480
gaucuuuc agcaggugau aguaccuag ucgacgggg uuucugccac gucaugugc	540
gggaucacca cggacuuggc gucuauuug gcguuucaga ggccgauucc gacaccacaa	600
caccgacagc agcaacaaca gcaaggcggc cuaucucua gccuuucucc ucagcuacaa	660
cagcaauua guuucauaa caauuuuua uccuacac caaggacaaa uauuguuacu	720
auuaggggaa cauauagug aaguucucg acaugguuu uaggcucua guaucugaaa	780
gcugcacaag agcuucuga ugaaguugu auuuuuuug gaaaaagca caaaggagau	840
gaucuuuaga aggaauuuc auugaauua gaaucuuug cuuuggcua ugauguaac	900
acuuauuguu cuggugugug ugaauuagc agcaggcaga aauuagagu ugcugugag	960
cuuacaacug cucaaagaca agaacuuaa augaaaaag ccaagcuuc ugcacugcu	1020
gaagaggug agcaaaggua cagacaguac caucaccaa ugcauuuuu uguuuuaua	1080
uuugagcaag uagcaggau uggaucagcc aaucuuuac cucauuuagc uuugcagca	1140
auuucgaagc auuucagau ccuaaaggau gcauuugcug agcauuuaa ggcgacgagc	1200
aagauuuag gugaagagga aggcuuugga gggaaaauc aaggcuuag acuuuuuu	1260

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guggaccauc aucuaaggca acaacgcgcg cugcaacaga uaggaaugau gcaaccaaau	1320
gcuuggagac cccaaaggagg uuuaaccugaa agagcugucu cuguccuucg ugcuuuggcuu	1380
uucgagcauu uucuucaucc uuacccaaag gauucagaca aaaucaugcu ugcuaagcaa	1440
acggggcuua caaggagcca ggugucuaac ugguucauaa augcucgagu ucgauuaugg	1500
aagccaaugg uagaagaaau guacuuggaa gaagugaaga aucaagaaca aaacaguacu	1560
aaacuucag gagauaacia aaacaagag accaaauaaa guguccaaa ugaagagaaa	1620
cauccaauua uuacuagcag cuuauuacia gaugguauua cuacuacua agcagaaaau	1680
ucuaccucaa cuauuucaac uuccccuacu gcaggugcuu cacuucaua ugcucacaa	1740
uucuccuucc uugguucauu caacauggau aaacuacua cuacuguuga ucauuuuga	1800
aaacacgcga aaaagcaaag aaugacaug cacaaguuuu cuccaaguag uauucuuua	1860
ucuguugaca uggaagccaa agcuagagaa ucaucaaaua agggguuuac uaaucuuua	1920
auggcagcau acgcgauggg agauuuugga agguuugauc cucaugauca acaauagacc	1980
gcgaauuuuc auggaaauaa ugguugucucu cuuacuuuag gacuuccucc uucugaaaac	2040
cuagccaugc cagugagcca acaaaauuac cuuucuaaug acuuuggaag uaggucugaa	2100
auggggaguc auuacaauag aaugggauau gaaaacauug auuuucagag ugggaauaag	2160
cgaauuccga cucaacuauu accagauuuu guuacaggua aucuaggaac augaauacca	2220
gaaagucucg uauugauagc ugaagagaa aaaggaaguu agggauacuc uuauuuug	2280
ugaggccuuc ugccccaggu cggaggaccc aaauugauac aaccuaucau aggagaaaag	2340
aaguggagac uaaaauaaag uaacaaaaau uuaaagcaca cuuucuaqua uauauacuuc	2400
uuuuuuuuu aguaagaaa agaagagauu ugugucuuu guguauagau agagucuauc	2460
uaguauaggu uauacuucua guuccuugag aagauugaua caacuaguag uauuuuuuu	2520
cuuuuggguu ggcuuggagu acuaauuuua guuauuggaa acuagcuaua guaaauug	2580
uaaaguugug auauuguucc ucucauuuug cauaauuuu gaaauuuu guaccuacua	2640
gcuaugucucu aaauuuguu uccauugcuu guaaugcaa uuuuuuuuga auuuuguc	2700
aucauuuuu gauuagcaaa aaaaaaaaaa aaaaa	2735

<210> SEQ ID NO 29
 <211> LENGTH: 72
 <212> TYPE: RNA
 <213> ORGANISM: Arabidopsis thaliana

<400> SEQUENCE: 29

aucagagugg cgcagcgga gcuugguggg cccauaacc acagguccca ggaucgaaac	60
cuggcucuga ua	72

<210> SEQ ID NO 30
 <211> LENGTH: 71
 <212> TYPE: RNA
 <213> ORGANISM: Arabidopsis thaliana

<400> SEQUENCE: 30

gcaccagugg ucuaguggca ugauaguacc cugccacggu acagaccgg guucaauucc	60
cggcuggugc a	71

<210> SEQ ID NO 31
 <211> LENGTH: 699
 <212> TYPE: DNA
 <213> ORGANISM: Cucurbita maxima

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<400> SEQUENCE: 31

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ggggcttcaa aagagaatta ggtcacctcc cagctcgggt tcgacacgcc atgccgagaa    60
atcgtgaccc tctagtcgtc gggagagtga tcggcgacgt cgtcgactcg ttctcgaggt    120
ccatctcgat tagggttggt tacgactcga gggaagttaa caatgggtgt gagctcaaac    180
cctctcaagc tgtcaacaag ccaagagttg agattggtgg cactgacctt cgcaccttct    240
tcactttggt tatggtggat cccgacgctc ctagccctag cgatcccaat ctaagagaat    300
acttgcaatt gttagtgaac gatattccag ctacaaccga ggcaaccttt ggacaagaga    360
tagtgtgcta cgagaatcca agaccaacgg tgggtatcca ccgttttggt ctggtcttgt    420
tccggcagct cggaaggcaa acggtgtatg ctccctgggtg gcgccagaac ttcaacacca    480
gacactttgc agagctttac aatcttggtt cggcagtcgc cgcgctctat ttcaattgcc    540
aaagggaaaa tggctccggt ggaaggagaa gagccggcga tgaatgttca taaaaacact    600
tcacttcaca ttatattatc aaccaatata ttgtaataac atggttcacg tttctatcta    660
atagattata tatttttaat aagttcgtga aaaaaaaaaa    699

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<210> SEQ ID NO 32

<211> LENGTH: 858

<212> TYPE: DNA

<213> ORGANISM: Cucurbita maxima

<400> SEQUENCE: 32

```

gagacaatta cgcactcttt cagctctctc acgtactacc atcctctcga cgccatgccg    60
agagaccgtg accctttggt cgttgggaga gtcacggcgc acgttatcga ctcggtcacg    120
aagtccattt cgattagggc tacttacaac aacagggaaa ttagcaatgg ctgtgagctc    180
aaacctctc aagttgtcaa ccagccaaga gttgagattg gtggcactga ctttcgcacc    240
ttcttcactt tggttatggt ggatcctgat gctcctagcc ctagtgatcc taatctaagg    300
gaatacttgc attggttggt gactgatatc ccagctacaa ctggagcgaa ctttggtcaa    360
gagatcgtgt gctatgagag cccaagaccc acggtgggta tccatcgtct tgtgctgggtg    420
ttgtttcgac agcttggaag gcaaacggtg tacgctcctg ggtggcgcca gaacttcaac    480
acaagagact ttgcagagct ttacaatctt ggcttgccgg tggcagccgt ttatttcaat    540
tgccaaaggg aaagtgggtc tggtggaagg agaagaacct aagatgatth ctaagcccca    600
cttcacatta attagattaa tattatagcc cctatcatct attaatccta ctttgctttt    660
agattaacct ttattttgag tacacccatg gatcataaat aagcccaaaa tgcattccta    720
atattgctct tatactcggt tcgtatgaat cactgtcttt tcttctttgt ttttcttggt    780
cgagtgttca tgtgtgctt ttttttctgt atgaatcaaa gtagaagatc aagattcgaa    840
aaaaaaaaaa aaaaaaaaaa    858

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<210> SEQ ID NO 33

<211> LENGTH: 596

<212> TYPE: DNA

<213> ORGANISM: Vitis vinifera

<400> SEQUENCE: 33

```

ccctcttgta ttgtatcggt gaggtgtgtg tgatgcctag ggaaagggat cctcttggtg    60
ttgggcgcgt tgcgggggat gttctggacc cctttctcag gtccatcact ctgaggggtga    120
cctacaataa tagagaagta gcaaatggct gtgagttcag accctctcag ctagtcagcc    180

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aacctagggt ggacattgga ggggatgact tgaggacctt ctatactttg gttatgggtg	240
acctgacgc tocaagcccc agtaatccga acctaaggga gtacttacat tggttggtga	300
ctgatattcc agcaactact ggggcaaaact tcggccaaga gattgtgtgt tatgagagcc	360
cacgccaac agctgggatt categctttg tttttgtatt gtttcgcca ctgggtaggc	420
agacagtgtg tgcaccaggg tggcgccaaa atttcaacac tagggacttt gctgagcttt	480
ataatcttgg tttgctgtt gctgctgttt attttaactg ccaaaggag ggcggtcgg	540
gtggtcgaag atcataatca atggattttg tacgcaacct tgcgacttac aaaggc	596

<210> SEQ ID NO 34

<211> LENGTH: 525

<212> TYPE: DNA

<213> ORGANISM: Malus domestica

<400> SEQUENCE: 34

atgcctaggg atagggaccc ccttgttgtt ggacgagtgg ttggtgatgt tttagacccc	60
ttcacaaggt ctgtttctct gaggtgacc tacggtacta aggaggtaa caatggttgt	120
gagctcaaac cttctgaagt tgccaacaa cctagagctg atattgttg agacgatctc	180
aggactttct acactctggt catggtggat cctgatgcac ccagccaag tgacccaac	240
ctaaaggaat atttgcattg gttggttacc gatattccag caactactgc ggcaagcttc	300
gggcaagaga tcgtgtgtta tgaaagtcca cggccaacag tggggattca tcgctttgtt	360
ttggtggtgt ttcgccaatt gggtaggcaa acggtgtatg ctccgggatg gcgccagaac	420
ttcaatacca gagacttcgc cgagctttat aatcttgat taccggtgtc tgcgtctat	480
tttaactgcc aaaggagggg cggtccggt ggaaggagaa gataa	525

<210> SEQ ID NO 35

<211> LENGTH: 745

<212> TYPE: DNA

<213> ORGANISM: Citrus unshiu

<400> SEQUENCE: 35

ggcagagga atagtcttac tacttttcta ggcgtgtgt gtatttgttt gtgcttagtg	60
ttggttgatg tttgtttgtg tttagtgttg ttgatatgtc tagcaggag agagatcctc	120
ttattgttgg ccgctgtgtt ggtgatgttc ttgacaattt tacaagaaca attccaatga	180
ggattaccta ttcaacaag gatgttaata atggccgtga gctcaaacct tctgaagtgc	240
tgaaccagcc tagggctgaa attggtggtg atgatcttag gacattttat actttggtaa	300
tggttgatcc tgatgcacca agcccaagtg accccagcct tagggagtat ttgcattggt	360
tggtgactga tattccagca accacagggg ccagctttgg ccaagagatt gtgaactatg	420
aaagccctag gccaacgatg gggattcaca ggtttgtctt tgtgtgttc cgccaacttg	480
ggaggcagac tgtttatgca ccagggtggc gtcagaactt cagcacgagg gattttgctg	540
agctttacaa tctgggacct ccggtggccg ctgtctactt caactgccag agggagagcg	600
gatccggcgg aaggcctgtc agacgatgat ccatacatgc ttaatttgat atcaaattac	660
acacacacac acacacacac acacacacac acacacacac actatttata	720
tatatatata tatatatata tatat	745

<210> SEQ ID NO 36

<211> LENGTH: 534

<212> TYPE: DNA

<213> ORGANISM: Lycopersicon esculentum

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<400> SEQUENCE: 36

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atgcctagag aacgtgatcc tcttgttgtt ggtcgtgttg taggggatgt attggaccct    60
ttcacaagaa ctattggcct aagagttata tatagagata gagaagtaa taatggatgc    120
gagcttaggc cttcccaagt tattaaccag ccaagggttg aagttggagg agatgaccta    180
cgtacctttt tcacttttgg tatggtggac cctgatgctc caagtccgag tgatccaaat    240
ctgagagaa accttcactg gttggtcacc gatattccag ctaccacagg ttcaagtttt    300
gggcaagaaa tagtgagcta tgaaagtcca agaccatcaa tgggaatata tcgatttgta    360
tttgtattat tcagacaatt aggtcggcaa acagtgtatg ctccaggatg gcgtcagaat    420
ttcaacacaa gagattttgc agaactttat aatcttggtt tacctgttgc tgctgtctat    480
tttaattgtc aaagagagag tggcagtggt ggacgtagaa gatctgctga ttga        534

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<210> SEQ ID NO 37

<211> LENGTH: 525

<212> TYPE: DNA

<213> ORGANISM: Populus tremula

<400> SEQUENCE: 37

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atgtcaaggg acagagatcc tctgagcgtt ggccgtgtta taggggacgt gctggacccc    60
ttcacaaagt ctatcccgtt cagggtcacc tacaactcca gagaggtaaa caatggttgc    120
gagctcaaac cctctcaggt tgccaaccag ccgaggggtg atattggcgg ggaagatcta    180
aggaccttct acactctggt tatggtggac cctgatgcac ccagcccaag tgacccacag    240
ctcagagaa atttgcattg gttggtgact gatattccag caacaacggg ggcaagcttt    300
ggccatgaaa ctgtgtgcta tgagagcccg aggcgcgaca tggggattca tcggtttgtt    360
ttcgtcttgt tccggcaact gggcaggcaa actgtgtatg cccctgggtg gcgccagaac    420
ttcaacacca gagactttgc tgaggctctac aatcttggtt cgcgggtggc tgctgtttat    480
ttcaactgcc agagggagag tggctctggt ggtaggaggc gataa        525

```

<210> SEQ ID NO 38

<211> LENGTH: 534

<212> TYPE: DNA

<213> ORGANISM: Hordeum vulgare

<400> SEQUENCE: 38

```

atggccggga gggacaggga tccgctggtt gtcggcaggg ttgtggggga cgtgctggac    60
cccttcgtcc gaaccaccaa cctcagggtg accttcggga acagggccgt gtccaacggc    120
tgcgagctca agccgtccat ggtcgcccag cagccgaggg tggaggtggg cggcaatgag    180
atgaggacct tctacacgct cgtgatggtg gaccagatg ctccaagtcc tagcgacccc    240
aaccttagag agtatctcca ctggttggtg acagatatcc cgggtacaac tggggcgctc    300
ttcgggcagg aggtgatgtg ctacgagagc cctcgtccaa ccatggggat ccaccgcttc    360
gtgctcgtgc tcttcagca gctggggcgg cagacggtgt acgccccggg gtggcgccag    420
aacttcaaca ccagggaact tgccgagctc tacaacctcg gccagcccggt tgccgccgct    480
tacttcaact gccagcgcca ggccggctcc ggccggcagga ggatgtacaa ttga        534

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<210> SEQ ID NO 39

<211> LENGTH: 537

<212> TYPE: DNA

<213> ORGANISM: Hordeum vulgare

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<400> SEQUENCE: 39

atggtgggga gcagcatgca gcgcggggac ccgctgggtg tggggcgggt gatcggcgac	60
gtggtggacc cgttcgtgcg gcgggtggcg ctgcgggtcg gctacgcgtc cagggaagtg	120
gccaacggct gcgagctccg gccgtccgcc atcgccgacc agccgcgcgt cgaggtcggc	180
ggcccggaaca tgcgcacctt ctacacctg gtgatgggtg atccggatgc tccaagcccc	240
agcgacccca gccttaggga gtacttgca tggctgggtc ccgacatccc ggccacgaca	300
ggagtgtctt ttgtaccga ggttgtgtgc tacgagggcc cgcggccggt gctcgggatc	360
caccgactgg tgttcctgct ctccagcaa ctgcggccgac agacggtgta cgcgccggg	420
tggcggcaga acttcagcac ccgcgacttt gccgagctct acaacctcgg cctgccgtc	480
gccgcgctct acttcaactg ccagaggag accggaaccg gcgggagaag gatgtga	537

<210> SEQ ID NO 40

<211> LENGTH: 847

<212> TYPE: DNA

<213> ORGANISM: *Oryza sativa*

<400> SEQUENCE: 40

tgcaccacac acagttcagc tagcagatca cctagctaga tagctgcctc taccacagta	60
tatttgctcc ctgcaacttg ctgctgctgc aatagctagc agctgcagct agtaagcaaa	120
actataaacc ttcagggttt ttgcaagat cgatggccgg aagtggcagg gacagggacc	180
ctcttggtgt tggtagggtt gtgggtgatg tgctggacgc gttcgtccgg agcaccaacc	240
tcaaggtcac ctatggctcc aagaccgtgt ccaatggctg cgagctcaag ccgtccatgg	300
tcaccacca gcctagggtc gaggtcggcg gcaatgacat gaggacattc tacacccttg	360
tgatggtaga ccagatgca ccaagcccaa gtgacctaa ccttagggag tatctacatt	420
ggttggtcac tgatattcct ggtactactg cagcgtcatt tgggcaagag gtgatgtgct	480
acgagagccc aaggccaacc atggggatcc accgctggt gttcgtgctg ttccagcagc	540
tggggcgtca gacagtgtac gcgcccggt ggcgtcagaa cttcaacacc aaggacttcg	600
ccgagctcta caacctcggc tcgccggctg ccgcgtcta cttcaactgc cagcgcgagg	660
caggctccgg cggcaggagg gtctaccct agctaacgat gatcccgatc gatctgctgc	720
atgctcacta tcacatcca gcatgtata cattgcaggt tcagacaatt gaaatgattc	780
tcgacacaca acatatatat gatggtgtaa ttaattatgc aattaaatag ctgagcaagg	840
ctaaggt	847

<210> SEQ ID NO 41

<211> LENGTH: 866

<212> TYPE: DNA

<213> ORGANISM: *Oryza sativa*

<400> SEQUENCE: 41

cctgtcactg tttggctagc ttaaccttc tgacatctat cctctggatt gaacggcagg	60
agatacctaa gctagctagc aatctctatc gatctgtttg ttacatgtt cagttaaagg	120
ttactgagaa atgcctagag tttttccggc tagcttcata agttagtggg ttagctgacc	180
tagattcaaa gtctaactct tttatttatt tgatattaga tatcctaacg tttttagtta	240
gaggttatta atttgacatg gccggcagcg gcagggacga tcctctgtg gttggcagga	300
ttgtgggtga tgtgctggat ccattcgtcc ggatcactaa cctcagtgtc agctatggtg	360
caaggatcgt ctccaatggc tgcgagctca agccgtccat ggtgacccaa cagcccagg	420

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tcgtgggtcgg tggcaatgac atgaggacgt tctacacact cgtgatggta gacccggatg	480
ctccgagccc aagcaaccct aaccttaggg agtatctaca ctggtcggtc accgatattc	540
ctggtaccac tggagcaaca tttgggcaag aggtgatgtg ctacgagagc ccaaggccaa	600
ccatggggat ccacgggtg gtgttcgtgc tgttcacga gctggggcgt cagacggtgt	660
acgcaccggg gtggcgccag aacttcagca ccaggaactt cgccgagctc tacaacctcg	720
gctcgccggt cgccaccgtc tacttcaact gccagcgga ggccggctcc ggcggcagga	780
gggtctaccc ctgctagct acgcatgcca cccggcctcc atgcatgcag cagctatagc	840
taagctgaga cctgcctagc tgtata	866

<210> SEQ ID NO 42
 <211> LENGTH: 848
 <212> TYPE: DNA
 <213> ORGANISM: Ipomoea nil

<400> SEQUENCE: 42

cacacacaca cacatatata tacagagaaa ggtagttagt gatcgaggag ctgagctagc	60
taggatgcga aggggaacag tagacccttt ggtgttgggg cgtgtgatcg gagacgttgt	120
ggatccattc acgaggtcg ttgagcttag ggtggtttac aataacgagg tggatatcag	180
gaatgggtgt gagatgagc cttctcagct catcaacca cctagggttg aaatcgccgg	240
acacgatctc cgtactttct aactctggt tatggtggat cctgatgctc caagtccaac	300
ctctccaacc ctgagggaa acctccactg gttggtcact gatataccag gaactacagg	360
agcaagcttc ggcaatgaag cgatattcta cgagcctcca agggcgtaaa tgggaatcca	420
ccgttttgtg tttgtgcttt tccggcaact tggccggcag acagtattatg caccggtttg	480
gcgccagaat ttcaacactc gaaactttgc tgagatttac aatcttggtt tgccagtggc	540
cgtcacttac ttaacggcc aaaggagggg tggcaccggc ggtcgatctc cggcagagcc	600
ctgggcagcc gattaattac cctgctcctt cccgttaatt tcatgcatgc atgcatgcta	660
tctatagcat aacatacata tagtatatat cataaataaa taagaccaca tgcattaaca	720
tgtttaattt tccatgaat atatgttaaa gttgttctag aagaactacg tactccatta	780
tattaccctt tatatatggc aatgaagatg gtttcactc tatttagaag ctaaaaaaaa	840
aaaaaaaa	848

<210> SEQ ID NO 43
 <211> LENGTH: 798
 <212> TYPE: DNA
 <213> ORGANISM: Arabidopsis thaliana

<400> SEQUENCE: 43

tttattgaga tacttgagat ccaagataaa tatgtcttta gtcgtagaga tcctcttgtg	60
gtcggcagtg ttgttgaga tgttcttgat ccttcacga ggttggctc tcttaaggtc	120
acttatggcc atagagaggt tactaatggc ttggatctaa ggccttctca agttctgaac	180
aaaccaatag tggagattgg aggagacgac ttcagaaatt tctacacctt ggttatggtg	240
gatccagatg tgccgagtc aagcaaccct caccaacgag aatatctcca ctggttggtg	300
actgatatac ctgccaccac tggaaatgcc tttggcaatg aggtgggtg ctacgagagt	360
ccacgtcccc cctcggaat tcatcgtatt gtgttggtat tgttcggca actcggaaga	420
caaacgggtt atgcaccggg gtggcgccaa cagttcaaca ctcgtgagtt tgctgagatc	480

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tacaatcttg gtcttctgt ggtgcctct tacttcaact gccagaggga gaatggctgt	540
gggggaagaa gaacgtagat gcgtacctac ttacgttaac taataatcta atcgtataat	600
attcccttaa tgaagtatgt aagcatctat gtcaatgtaa taagaattta aagatacgag	660
ctaaaaaaaa tgatgcatat gctgacatcg atgtaaagta gtttacctt ttaatgtaat	720
aactagggtt taacccgcgg tacaccgca gactattttg tttttttaag aataaaaaata	780
taatttggtt agtcgatt	798

<210> SEQ ID NO 44
 <211> LENGTH: 519
 <212> TYPE: DNA
 <213> ORGANISM: Glycine max

<400> SEQUENCE: 44

atggcacggg agaaccctct tggtatttgt ggtgtgattg gggatgttct caaccctttt	60
acaagctccg tttctttgac tggttcaatc aataaatagg cgattagcaa tggcttgga	120
ctcaggccct ctcaagtgtg taatcgccct agggttactg ttggtggtga agacctagg	180
accttctaca ctctgggtat ggtggatgca gatgcacctc gccctagcaa cctgtcttg	240
aggaataacc ttcactggat ggtgacagat attccagcta ccacaaatgc aagctttggg	300
agagaggttg tggtttatga gagccgaac ctttcagtag ggattcatcg aatcgtgttc	360
gtattgttcc agcaattggg cagagacact gtcacacccc cagaatggcg ccataatttc	420
aattccagaa actttgctga aattaataac cttgcacctg ttgcagcagc ttatgccaac	480
tgccaaagag agcgtggttg cggtggaagg agatattaa	519

<210> SEQ ID NO 45
 <211> LENGTH: 560
 <212> TYPE: DNA
 <213> ORGANISM: Zea mays

<400> SEQUENCE: 45

agagcacatc cgtagtgtgt gcatgcatca cagtcacaca cacacagcag aagaagaaga	60
aaccgaacga ggggttagct agcaaaataa acagaagcaa gcaagctagc tagagctaag	120
gatcgagatc gagatcgacc gaccgacgac gatcagctag catggcgcg ctcgtggatc	180
cgctgggtgt gggggcgggt atcgccgagg tgggtggacct gttcgtgcct tccatctcca	240
tgaccgtcgc ctatgatggc cccaaggaca tcagcaacgg ctgcctcctc aagccgtccg	300
ccaccgccgc gcccgccctc gtcgcgcatc ccggccgcgc caacgacctc tacacgtga	360
tcatgacgga ccccgatgcg cctagcccca gcaacccgac catgagggag tacctccact	420
ggatagtgat taacatacca ggaggaacag atgctactaa aggtgaggag gtggtggagt	480
acatgggccc gcggccgcgc gtgggtatcc accgctacgt gctgggtgctg ttcgagcaga	540
agacgcgcgt gcacgaggag	560

<210> SEQ ID NO 46
 <211> LENGTH: 1069
 <212> TYPE: DNA
 <213> ORGANISM: Zea mays

<400> SEQUENCE: 46

tggaaaaaac ccagcgcttt gtgccgccgc cgtccgccgc cccctctgcc cttgtacgcg	60
cacctagaca catcgtcatc gatcatcaca cgcaatcgac acaagaagtt aataaacagc	120
ccaaggacgc agagatcagc tgatcgagaa ggacttgtac tactactcag tattgtcgtc	180

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acatgcacat atatgtacat aaagagctag ctacctgagc tctaccaag gtcgcgttga	240
tcgatcgatc atggcgcggt tcgtggaccc gctggtggtg gggcgggtga tcggcgaggt	300
ggtggacctg ttcgtgcctt ccgtctccat gaccgtcgcc tatggcccca aagacatcag	360
caacggctgc ctctcaagc cgtccgccac cgcgcgcgcg ccgctcgccc gcctctccgg	420
ccgcgcgcac gacctctaca cgctgatcat gacggaccca gatgcgccta gcccagcgga	480
cccgacctag agggagtagc tccactggat agtgactaac ataccaggag gaacggatgc	540
aaacaaaggt gaggaggtg tggagtacat gggccgcgcg ccgccggtcg gaatccaccg	600
ctacgtgctg gtgtgttctg agcagaagac gcgtgtgcac gcggagggtc ccggtgagcg	660
cgccaacttc aacacacgcg cgttcgcggc ggccgcacgag ctccggcctcc ccaccgccgt	720
cgtgtacttc aacgcgcaga aagagccggc caaccaccgc cgcgcgtagc tagtagctcc	780
aaacagggcg cgcagctga gctgcgtgcg tgcaaccac cacacagccg ccggcgaagg	840
ctgcctatat gaccggcgaa taaaaagtct tactgcaccg tccgtaagcg tactctctgt	900
tggtatatgc ttgtcttcag gctcttgagt ctatctactt aaatgtggtt accactgagt	960
aatagaagca gttggcgctt cgatcgatca ttctaataac cgtacgtgtc aatcattcct	1020
gtttccatca tcttgcatth gaagacgcat tggttctaca ccaaggtgt	1069

<210> SEQ ID NO 47

<211> LENGTH: 1288

<212> TYPE: DNA

<213> ORGANISM: Cucurbita maxima

<400> SEQUENCE: 47

gactttttat tcaacaatct ctctctctct ctctcaactt ccgatcaagt ctctccgccg	60
tcttttcacc ggagctgaca attccgatca ttttttgctt cccttaaatt tccggcatgg	120
aggaaccacc gccaaacgcc ttggatttgc cccctggctt cagattccac ccacccgacg	180
aggagatcgt cacttattac ctgatacata agatcaccga cgcgccttc actgccaccg	240
ccatcgagga agctgacctg aataagtgtg aaccttggga tttgccacat aaagctaaga	300
tgggggaaaa agaatggtat tttttttgcc agagagaccg gaaatatccg accgggatga	360
gaacgaaccg ggcgactcag accggttact ggaaagcgac cgggaagac aaggagattc	420
tcaagggaag aacggttctg gctggtatga agaaaacgct ggttttttac aaagggaag	480
ctcccaaaag tgaaaagacc aattgggtca tgcctgaatt tcgactcgaa cccaaattct	540
ttcagtttct tggttttccc aagccatta aggetgattg ggttgatgt cggttttttc	600
acaagaacac aacgaacacg gtcggagtag tgaaaaagat tcaaacttct gatttttctt	660
cttctctccc acctctaata gatccacaa ctgctcatac tccaatcagt ggcagattcg	720
ataatggtga agtcaactgg aggttatcag taccattcga taattatgca aatgattacc	780
attatcatcg gcctttttca gcgacgaata ctgcagtgc aatgatttcg tcgtacccat	840
cgtctgtccc cgacgacgaa ttcttctcat ttgatcaact agacgtcggt ggaacaatgt	900
caatggcggc ggccgacgaca acaacaacaa caactatgga gtgcaaaata gaacaagttt	960
catggtcaac gatgagcggt gtgacaccg agatatcgc gtcgattgac aacgaggcag	1020
ctctcgagtt ctgggactac tgaaaattga aagtagatgt tatgatcgaa caatggcgat	1080
gctttgtttt aaatgggcat ttcccatatt gaacgtttta acaatgatta attgattgct	1140
aattattatt attttttttt tttggttaca tagtcctttt tgggaaggaa tattagaact	1200

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ttcatgggtt tggtttggtg attgtattga tatgtagcaa tgtgacattg tatatagctt	1260
ctttatcttt tattttaacc gttgcaaa	1288
 <210> SEQ ID NO 48	
<211> LENGTH: 1964	
<212> TYPE: DNA	
<213> ORGANISM: Arabidopsis thaliana	
 <400> SEQUENCE: 48	
taataatcat ttttttctt ataaccttc tctctatctt tacaatttat ttgttatta	60
gaagtggtag tggagtga aaacaaatcc taagcagtc taaccgatcc ccgaagctaa	120
agattcttca ccttccaaa taaagcaaaa cctagatccg acattgaagg aaaaacctt	180
tagatccatc tctgaaaaa aaccaaccat gaagagagat catcatcatc atcatcaaga	240
taagaagact atgatgatga atgaagaaga cgacggtaac ggcatggatg agcttctagc	300
tgttcttggt tacaagggtt ggtcatcgga aatggctgat gttgctcaga aactcgagca	360
gcttgaagtt atgatgtcta atgttcaaga agacgatctt tctcaactcg ctactgagac	420
tgttcactat aatcggcggg agctttacac gtggcttgat tctatgctca ccgaccttaa	480
tcctccgtcg tctaacgccc agtacgatct taaagctatt cccggtgacg cgattctcaa	540
tcagttcgct atcgattcgg cttcttcgtc taaccaaggc ggcggaggag atacgtatac	600
tacaaacaag cgggtgaaat gctcaaacgg cgtcgtggaa accaccacag cgacggctga	660
gtcaactcgg catgttgctc tggttgactc gcaggagaac ggtgtgcgtc tegtacacgc	720
gcttttggtc tgcgctgaag ctgttcagaa ggagaatctg actgtggcgg aagctctggt	780
gaagcaaato ggattcttag ctgtttctca aatcggagct atgagaaaag tcgctactta	840
cttcgcccga gctctcggc ggcggattta ccgtctctct ccgtcgcaga gtccaatcga	900
ccactctctc tccgatactc ttcagatgca cttctacgag acttgctcctt atctcaagtt	960
cgctcacttc acggcgaatc aagcgattct cgaagctttt caagggaaga aaagagtcca	1020
tgtcattgat ttctctatga gtcaagggtc tcaatggccg gcgcttatgc aggtcttgct	1080
gcttcgacct ggtggctctc ctgttttccg gtttaaccga attggctccac cggcaccgga	1140
taatttcgat tatcttcatg aagttgggtg taagctggct catttagctg aggcgattca	1200
cgttgagttt gactacagag gatttggtgc taacacttta gctgatcttg atgcttcgat	1260
gcttgagctt agaccaagtg agattgaatc tgttgcggtt aactctgttt tcgagcttca	1320
caagctcttg ggacgacctg gtgcgacga taaggttctt ggtgtggtga atcagattaa	1380
accggagatt ttcactgttg ttgagcagga atcgaaccat aatagtcga tttctttaga	1440
tcggtttact gagtcgttgc attattactc gacgttggtt gactcgttgg aaggtgtacc	1500
gagtggtcaa gacaagggtc tgtcggaggt ttacttgggt aaacagatct gcaacgttgt	1560
ggcttggtgat ggacctgacc gagttgagcg tcatgaaacg ttgagtcagt ggaggaaaccg	1620
gttcgggtct gctgggtttg cggctgcaca tattgggttcg aatgcgttta agcaagcgag	1680
tatgcttttg gctctgttca acggcgggtg gggttatcgg gtggaggaga gtgacggctg	1740
tctcatgttg ggttggcaca caccgacgct catagccacc tcggcttgga aactctccac	1800
caattagatg gtggctcaat gaattgatct gttgaaccgg ttatgatgat agatttcga	1860
ccgaagccaa actaaactct actgttttct cctttgtcac ttgttaagat cttatcttct	1920
attatattag gtaattgaaa aatttctaaa ttactcacac tggc	1964

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<210> SEQ ID NO 49
 <211> LENGTH: 1556
 <212> TYPE: DNA
 <213> ORGANISM: Lycopersicon esculentum

<400> SEQUENCE: 49

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aaagaaaaaa ggaatattgt gtgtttgctt ttttttctga ctagtagtat tgctaactat    60
gtattccatt aaggatttgc tgtgaaaaag cctgatatca gtaagcataa aactcgggag    120
atcacttaca cacacacaca cctcctctaaa aaagagaaga gagatttact gttaaacaga    180
ggtttttttc catttctttt ttttttctag tgtgtgtgtg agagaaagag atgattttca    240
taggcacaaa caaatagaaa ggaacaaaat ttagagttaa gaagaaagtg tgtgagagaa    300
taatggaggg tggttctagt ggaataacta gtacatcttg ttaaatgatg atgggatatg    360
gagatcatga aaacaacaac aacaacaatg gaaatggtaa tggaaatgga aatggaaatg    420
taacaatttg tgctctcca atgatgatga tgatgctcc tctcctcct tctttaacta    480
acaataacaa tgcagaaaca agcaacaaca acatcctttt tctccttctc atggacaaca    540
acaacaacaa taatctcaaa gaagacaaca actcttcttc tcttccatc aagtcaaaga    600
ttatggctca tctcactac catcgtctct tgactgctta tctcaattgt caaaagatag    660
gagctccgcc agaagtgggt gcaaggctag aggaaatatg tgccacgtca gcaacaatgg    720
gccgtagcag tagtagtagt ggtggtgtaa tcattggaga agatcctgca ctagatcagt    780
tcattggagg ttattgtgag atgctgacaa aatatgaaca agaactctca aaacccttca    840
aggaagccat ggtttttctt tcaagaattg agtgtcagtt caaagcttta actcttgcac    900
ctaattcttc tcataatct gctttgggct aggcaatgga tagaaatgga tcactgatg    960
aagaggttga cgtgaataac agtttcatcg acccccaggc tgaggataga gagtcaaag   1020
gtcaattgtt gcgtaagtac agcgggtact tgggaagcct taagcaggag ttcataaaga   1080
agaggaagaa aggcaagctg cctaaggaag caaggcaaca attggtggat tgggtggctta   1140
gacatattaa atggccatat ccatcggaat ctcagaagct tgcaactagct gaatcaacgg   1200
gattggacca gaagcaaaata aacaactggt ttatcaatca aagaaagagg cattggaaac   1260
catcagaaga tatgcagttt gttgtgatgg atgctgctca tccacattac tatatggata   1320
atgttcttgc taaccatttc ccaatggata tgacaccctc tctcctctga attaagattt   1380
gtcattatta atatcaagga tgtttaatta atttgcatat tacttgtgtg catgtagtag   1440
tacaagctat tgtgacacaa tcaacttttt attagaccaa atatataaag tgcttgaat   1500
agatctttct attatcatct ttaattatgg aattaaatag tttgtacttg ctaaaa    1556

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<210> SEQ ID NO 50
 <211> LENGTH: 2735
 <212> TYPE: DNA
 <213> ORGANISM: Solanum tuberosum

<400> SEQUENCE: 50

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catgcagaga taaaaatata gatcagcttg acaagaaggc aacttctcaa agcttagaga    60
gtaccacccc gaagatagac agttagttac atgtactgtt atagataaaa ggagaaatcc   120
gaagaagaaa gaattttttt tgcagatatg tactatcaag gaacctcgga taataactaat   180
atacaagctg atcatcaaca acgtcataat catgggaata gtaataataa taatattcag   240
acactttatt tgatgaaccc taacaattat atgcaaggct acactacttc tgacacacag   300
cagcagcagc agttactttt cctgaattct tcaccagcag caagcaacgc gctttgccat   360

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gcgaatatac aacacgcgcc gctgcaacag cagcactttg tccgtgtgcc tcttcggca	420
gtaagtttgc acgatcagat caatcatcat ggacttttac agcgcagtgtg gaacaaccaa	480
gatcaatctc agcagggtgat agtaccatcg tcgacggggg tttctgccac gtcagtgtgc	540
gggatcacca cggacttggc gtctcaattg gcgtttcaga ggccgattcc gacaccacaa	600
caccgacagc agcaacaaca gcaaggcggg ctatctctaa gcctttctcc tcagctacaa	660
cagcaaatta gtttcaataa caatatttca tctcatcac caaggacaaa taatgttact	720
attaggggaa cattagatgg aagttctagc aacatggttt taggctctaa gtatctgaaa	780
gctgcacaag agcttcttga tgaagtgtt aatattgttg gaaaagcat caaaggagat	840
gatcaaaaga aggataattc aatgaataaa gaatcaatgc ctttggctag tgatgtcaac	900
actaatagtt ctggtggtgg tgaaagtagc agcaggcaga aaaatgaagt tgctgttgag	960
cttacaactg ctcaaagaca agaactcaa atgaaaaaag ccaagcttct tgccatgctt	1020
gaagaggtgg agcaaaggta cagacagtag catcaccaaa tgcaataat tgtattatca	1080
tttgagcaag tagcaggaat tggatcagcc aaatcataca ctcaattagc tttgcatgca	1140
atttcgaagc aattcagatg cctaaaggat gcaattgctg agcaagtaa ggcgacgagc	1200
aagagtttag gtgaagagga aggcttggga gggaaaatcg aaggctcaag actcaaattt	1260
gtggaccatc atctaaggca acaacgcgcg ctgcaacaga taggaatgat gcaaccaa	1320
gcttggagac ccaaagagg tttacctgaa agagctgtct ctgtccttcg tgcttggctt	1380
ttcgagcatt ttctcatcc ttaccctaaag gattcagaca aaatcatgct tgctaagcaa	1440
acggggctaa caaggagcca ggtgtctaac tggttcataa atgctcgagt tcgattatgg	1500
aagccaatgg tagaagaat gtacttgaa gaagtgaaga atcaagaaca aaacagtact	1560
aatacttcag gagataacaa aaacaagag accaatataa gtgctccaaa tgaagagaaa	1620
catccaatta ttactagcag ctattacaa gatggtatta ctactactca agcagaaatt	1680
tctacctcaa ctatttcaac tccccctact gcagggtgctt cacttcatca tgctcacaat	1740
ttctccttcc ttggttcatt caacatggat aatactacta ctactgttga tcatattgaa	1800
aacaacgcga aaaagcaaag aaatgacatg cacaagtttt ctccaagtag tattctttca	1860
tctgttgaca tggaagccaa agctagagaa tcatcaataa aagggtttac taatccttta	1920
atggcagcat acgcgatggg agattttgga aggtttgatc ctcatgatca acaaatgacc	1980
gcgaattttc atggaataa tgggtgtctct ctacttttag gacttcctcc ttctgaaaac	2040
ctagccatgc cagtgagcca acaaaattac ctttctaatag acttgggaag taggtctgaa	2100
atggggagtc attacaatag aatgggatat gaaaacattg attttcagag tgggaataag	2160
cgatttccga ctcaactatt accagatttt gttacaggtg atctaggaac atgaatacca	2220
gaaagtctcg tattgatagc tgaaaagata aaaggaagtt agggatactc ttatattgtg	2280
tgaggccttc tggcccaagt cggaggaccc aatttgatac aacctatcat aggagaaaag	2340
aagtggagac taaattaaag taacaaaatt ttaaagcaca ctttctagta tatatacttc	2400
ttttttttat agtatagaaa agaagagatt ttgtgcttta gtgtatagat agagtctact	2460
tagtataggt tatacttcta gttccttgag aagattgata caactagtag tatttttttt	2520
cttttgggtt ggcttggagt actattttta gttatttgaa actagctata gtaaatgttg	2580
taaagttgtg atattgttcc tctcaatttg catataattt gaaatatttt gtacctacta	2640
gctagtctct aaattatgtt tccattgctt gtaattgcaa ttttatttga attttgtgct	2700
atcattatta gattagcaaa aaaaaaaaaa aaaaa	2735

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<210> SEQ ID NO 51
 <211> LENGTH: 72
 <212> TYPE: DNA
 <213> ORGANISM: Arabidopsis thaliana

<400> SEQUENCE: 51

atcagagtgg cgcagcggaa gcgtggtggg cccataaccc acaggtccca ggatcgaaac 60
 ctggctctga ta 72

<210> SEQ ID NO 52
 <211> LENGTH: 71
 <212> TYPE: DNA
 <213> ORGANISM: Arabidopsis thaliana

<400> SEQUENCE: 52

gcaccagtgg tctagtggca tgatagtacc ctgccacggt acagaccgg gttcaattcc 60
 cggctggtgc a 71

<210> SEQ ID NO 53
 <211> LENGTH: 1230
 <212> TYPE: PRT
 <213> ORGANISM: Francisella tularensis

<400> SEQUENCE: 53

Ile Ser Glu Asp Leu Leu Gln Asn Tyr Ser Asp Val Tyr Phe Lys Leu
 1 5 10 15
 Lys Lys Ser Asp Asp Asp Asn Leu Gln Lys Asp Phe Lys Ser Ala Lys
 20 25 30
 Asp Thr Ile Lys Lys Gln Ile Ser Glu Tyr Ile Lys Asp Ser Glu Lys
 35 40 45
 Phe Lys Asn Leu Phe Asn Gln Asn Leu Ile Asp Ala Lys Lys Gly Gln
 50 55 60
 Glu Ser Asp Leu Ile Leu Trp Leu Lys Gln Ser Lys Asp Asn Gly Ile
 65 70 75 80
 Glu Leu Phe Lys Ala Asn Ser Asp Ile Thr Asp Ile Asp Glu Ala Leu
 85 90 95
 Glu Ile Ile Lys Ser Phe Lys Gly Trp Thr Thr Tyr Phe Lys Gly Phe
 100 105 110
 His Glu Asn Arg Lys Asn Val Tyr Ser Ser Asn Asp Ile Pro Thr Ser
 115 120 125
 Ile Ile Tyr Arg Ile Val Asp Asp Asn Leu Pro Lys Phe Leu Glu Asn
 130 135 140
 Lys Ala Lys Tyr Glu Ser Leu Lys Asp Lys Ala Pro Glu Ala Ile Asn
 145 150 155 160
 Tyr Glu Gln Ile Lys Lys Asp Leu Ala Glu Glu Leu Thr Phe Asp Ile
 165 170 175
 Asp Tyr Lys Thr Ser Glu Val Asn Gln Arg Val Phe Ser Leu Asp Glu
 180 185 190
 Val Phe Glu Ile Ala Asn Phe Asn Asn Tyr Leu Asn Gln Ser Gly Ile
 195 200 205
 Thr Lys Phe Asn Thr Ile Ile Gly Gly Lys Phe Val Asn Gly Glu Asn
 210 215 220
 Thr Lys Arg Lys Gly Ile Asn Glu Tyr Ile Asn Leu Tyr Ser Gln Gln
 225 230 235 240
 Ile Asn Asp Lys Thr Leu Lys Lys Tyr Lys Met Ser Val Leu Phe Lys

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245								250					255				
Gln	Ile	Leu	Ser	Asp	Thr	Glu	Ser	Lys	Ser	Phe	Val	Ile	Asp	Lys	Leu		
		260						265					270				
Glu	Asp	Asp	Ser	Asp	Val	Val	Thr	Thr	Met	Gln	Ser	Phe	Tyr	Glu	Gln		
		275					280					285					
Ile	Ala	Ala	Phe	Lys	Thr	Val	Glu	Glu	Lys	Ser	Ile	Lys	Glu	Thr	Leu		
	290					295					300						
Ser	Leu	Leu	Phe	Asp	Asp	Leu	Lys	Ala	Gln	Lys	Leu	Asp	Leu	Ser	Lys		
305				310						315					320		
Ile	Tyr	Phe	Lys	Asn	Asp	Lys	Ser	Leu	Thr	Asp	Leu	Ser	Gln	Gln	Val		
			325						330					335			
Phe	Asp	Asp	Tyr	Ser	Val	Ile	Gly	Thr	Ala	Val	Leu	Glu	Tyr	Ile	Thr		
		340						345					350				
Gln	Gln	Ile	Ala	Pro	Lys	Asn	Leu	Asp	Asn	Pro	Ser	Lys	Lys	Glu	Gln		
		355					360					365					
Glu	Leu	Ile	Ala	Lys	Lys	Thr	Glu	Lys	Ala	Lys	Tyr	Leu	Ser	Leu	Glu		
	370					375					380						
Thr	Ile	Lys	Leu	Ala	Leu	Glu	Glu	Phe	Asn	Lys	His	Arg	Asp	Ile	Asp		
385					390					395					400		
Lys	Gln	Cys	Arg	Phe	Glu	Glu	Ile	Leu	Ala	Asn	Phe	Ala	Ala	Ile	Pro		
				405					410					415			
Met	Ile	Phe	Asp	Glu	Ile	Ala	Gln	Asn	Lys	Asp	Asn	Leu	Ala	Gln	Ile		
			420					425					430				
Ser	Ile	Lys	Tyr	Gln	Asn	Gln	Gly	Lys	Lys	Asp	Leu	Leu	Gln	Ala	Ser		
		435					440					445					
Ala	Glu	Asp	Asp	Val	Lys	Ala	Ile	Lys	Asp	Leu	Leu	Asp	Gln	Thr	Asn		
	450					455					460						
Asn	Leu	Leu	His	Lys	Leu	Lys	Ile	Phe	His	Ile	Ser	Gln	Ser	Glu	Asp		
465					470					475					480		
Lys	Ala	Asn	Ile	Leu	Asp	Lys	Asp	Glu	His	Phe	Tyr	Leu	Val	Phe	Glu		
			485						490					495			
Glu	Cys	Tyr	Phe	Glu	Leu	Ala	Asn	Ile	Val	Pro	Leu	Tyr	Asn	Lys	Ile		
			500					505					510				
Arg	Asn	Tyr	Ile	Thr	Gln	Lys	Pro	Tyr	Ser	Asp	Glu	Lys	Phe	Lys	Leu		
		515					520					525					
Asn	Phe	Glu	Asn	Ser	Thr	Leu	Ala	Asn	Gly	Trp	Asp	Lys	Asn	Lys	Glu		
	530					535					540						
Pro	Asp	Asn	Thr	Ala	Ile	Leu	Phe	Ile	Lys	Asp	Asp	Lys	Tyr	Tyr	Leu		
545					550					555					560		
Gly	Val	Met	Asn	Lys	Lys	Asn	Asn	Lys	Ile	Phe	Asp	Asp	Lys	Ala	Ile		
			565						570					575			
Lys	Glu	Asn	Lys	Gly	Glu	Gly	Tyr	Lys	Lys	Ile	Val	Tyr	Lys	Leu	Leu		
			580					585					590				
Pro	Gly	Ala	Asn	Lys	Met	Leu	Pro	Lys	Val	Phe	Phe	Ser	Ala	Lys	Ser		
		595					600					605					
Ile	Lys	Phe	Tyr	Asn	Pro	Ser	Glu	Asp	Ile	Leu	Arg	Ile	Arg	Asn	His		
	610					615					620						
Ser	Thr	His	Thr	Lys	Asn	Gly	Ser	Pro	Gln	Lys	Gly	Tyr	Glu	Lys	Phe		
625					630					635					640		
Glu	Phe	Asn	Ile	Glu	Asp	Cys	Arg	Lys	Phe	Ile	Asp	Phe	Tyr	Lys	Gln		
			645						650					655			
Ser	Ile	Ser	Lys	His	Pro	Glu	Trp	Lys	Asp	Phe	Gly	Phe	Arg	Phe	Ser		
		660						665					670				

Asp	Thr	Gln	Arg	Tyr	Asn	Ser	Ile	Asp	Glu	Phe	Tyr	Arg	Glu	Val	Glu	
	675						680					685				
Asn	Gln	Gly	Tyr	Lys	Leu	Thr	Phe	Glu	Asn	Ile	Ser	Glu	Ser	Tyr	Ile	
	690					695					700					
Asp	Ser	Val	Val	Asn	Gln	Gly	Lys	Leu	Tyr	Leu	Phe	Gln	Ile	Tyr	Asn	
705					710						715				720	
Lys	Asp	Phe	Ser	Ala	Tyr	Ser	Lys	Gly	Arg	Pro	Asn	Leu	His	Thr	Leu	
				725					730					735		
Tyr	Trp	Lys	Ala	Leu	Phe	Asp	Glu	Arg	Asn	Leu	Gln	Asp	Val	Val	Tyr	
			740					745					750			
Lys	Leu	Asn	Gly	Glu	Ala	Glu	Leu	Phe	Tyr	Arg	Lys	Gln	Ser	Ile	Pro	
							760					765				
Lys	Lys	Ile	Thr	His	Pro	Ala	Lys	Glu	Ala	Ile	Ala	Asn	Lys	Asn	Lys	
	770						775					780				
Asp	Asn	Pro	Lys	Lys	Glu	Ser	Val	Phe	Glu	Tyr	Asp	Leu	Ile	Lys	Asp	
785					790						795				800	
Lys	Arg	Phe	Thr	Glu	Asp	Lys	Phe	Phe	Phe	His	Cys	Pro	Ile	Thr	Ile	
				805					810					815		
Asn	Phe	Lys	Ser	Ser	Gly	Ala	Asn	Lys	Phe	Asn	Asp	Glu	Ile	Asn	Leu	
								825					830			
Leu	Leu	Lys	Glu	Lys	Ala	Asn	Asp	Val	His	Ile	Leu	Ser	Ile	Asp	Arg	
		835					840					845				
Gly	Glu	Arg	His	Leu	Ala	Tyr	Tyr	Thr	Leu	Val	Asp	Gly	Lys	Gly	Asn	
	850					855						860				
Ile	Ile	Lys	Gln	Asp	Thr	Phe	Asn	Ile	Ile	Gly	Asn	Asp	Arg	Met	Lys	
865					870					875					880	
Thr	Asn	Tyr	His	Asp	Lys	Leu	Ala	Ala	Ile	Glu	Lys	Asp	Arg	Asp	Ser	
				885					890					895		
Ala	Arg	Lys	Asp	Trp	Lys	Lys	Ile	Asn	Asn	Ile	Lys	Glu	Met	Lys	Glu	
				900				905					910			
Gly	Tyr	Leu	Ser	Gln	Val	Val	His	Glu	Ile	Ala	Lys	Leu	Val	Ile	Glu	
		915					920						925			
Tyr	Asn	Ala	Ile	Val	Val	Phe	Glu	Asp	Leu	Asn	Phe	Gly	Phe	Lys	Arg	
	930					935					940					
Gly	Arg	Phe	Lys	Val	Glu	Lys	Gln	Val	Tyr	Gln	Lys	Leu	Glu	Lys	Met	
945					950					955					960	
Leu	Ile	Glu	Lys	Leu	Asn	Tyr	Leu	Val	Phe	Lys	Asp	Asn	Glu	Phe	Asp	
				965					970					975		
Lys	Thr	Gly	Gly	Val	Leu	Arg	Ala	Tyr	Gln	Leu	Thr	Ala	Pro	Phe	Glu	
			980						985				990			
Thr	Phe	Lys	Lys	Met	Gly	Lys	Gln	Thr	Gly	Ile	Ile	Tyr	Tyr	Val	Pro	
		995					1000					1005				
Ala	Gly	Phe	Thr	Ser	Lys	Ile</										

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Phe	Arg	Asn	Ser	Asp	Lys	Asn	His	Asn	Trp	Asp	Thr	Arg	Glu	Val
1085						1090					1095			
Tyr	Pro	Thr	Lys	Glu	Leu	Glu	Lys	Leu	Leu	Lys	Asp	Tyr	Ser	Ile
1100						1105					1110			
Glu	Tyr	Gly	His	Gly	Glu	Cys	Ile	Lys	Ala	Ala	Ile	Cys	Gly	Glu
1115						1120					1125			
Ser	Asp	Lys	Lys	Phe	Phe	Ala	Lys	Leu	Thr	Ser	Val	Leu	Asn	Thr
1130						1135					1140			
Ile	Leu	Gln	Met	Arg	Asn	Ser	Lys	Thr	Gly	Thr	Glu	Leu	Asp	Tyr
1145						1150					1155			
Leu	Ile	Ser	Pro	Val	Ala	Asp	Val	Asn	Gly	Asn	Phe	Phe	Asp	Ser
1160						1165					1170			
Arg	Gln	Ala	Pro	Lys	Asn	Met	Pro	Gln	Asp	Ala	Asp	Ala	Asn	Gly
1175						1180					1185			
Ala	Tyr	His	Ile	Gly	Leu	Lys	Gly	Leu	Met	Leu	Leu	Gly	Arg	Ile
1190						1195					1200			
Lys	Asn	Asn	Gln	Glu	Gly	Lys	Lys	Leu	Asn	Leu	Val	Ile	Lys	Asn
1205						1210					1215			
Glu	Glu	Tyr	Phe	Glu	Phe	Val	Gln	Asn	Arg	Asn	Asn			
1220						1225					1230			

<210> SEQ ID NO 54
 <211> LENGTH: 38
 <212> TYPE: RNA
 <213> ORGANISM: Francisella tularensis

<400> SEQUENCE: 54

gggucuaaga acuuuaaaaua auuucuacug uuguacau

38

<210> SEQ ID NO 55
 <211> LENGTH: 1228
 <212> TYPE: PRT
 <213> ORGANISM: Unknown
 <220> FEATURE:
 <223> OTHER INFORMATION: Lachnospiraceae bacterium

<400> SEQUENCE: 55

Ala	Ala	Ser	Lys	Leu	Glu	Lys	Phe	Thr	Asn	Cys	Tyr	Ser	Leu	Ser	Lys
1				5					10					15	
Thr	Leu	Arg	Phe	Lys	Ala	Ile	Pro	Val	Gly	Lys	Thr	Gln	Glu	Asn	Ile
		20						25					30		
Asp	Asn	Lys	Arg	Leu	Leu	Val	Glu	Asp	Glu	Lys	Arg	Ala	Glu	Asp	Tyr
		35				40						45			
Lys	Gly	Val	Lys	Lys	Leu	Leu	Asp	Arg	Tyr	Tyr	Leu	Ser	Phe	Ile	Asn
	50				55						60				
Asp	Val	Leu	His	Ser	Ile	Lys	Leu	Lys	Asn	Leu	Asn	Asn	Tyr	Ile	Ser
65					70					75				80	
Leu	Phe	Arg	Lys	Lys	Thr	Arg	Thr	Glu	Lys	Glu	Asn	Lys	Glu	Leu	Glu
			85						90					95	
Asn	Leu	Glu	Ile	Asn	Leu	Arg	Lys	Glu	Ile	Ala	Lys	Ala	Phe	Lys	Gly
		100						105					110		
Ala	Ala	Gly	Tyr	Lys	Ser	Leu	Phe	Lys	Lys	Asp	Ile	Ile	Glu	Thr	Ile
		115					120						125		
Leu	Pro	Glu	Ala	Ala	Asp	Asp	Lys	Asp	Glu	Ile	Ala	Leu	Val	Asn	Ser
	130					135						140			
Phe	Asn	Gly	Phe	Thr	Thr	Ala	Phe	Thr	Gly	Phe	Phe	Asp	Asn	Arg	Glu
145					150					155					160

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Asn Met Phe Ser Glu Glu Ala Lys Ser Thr Ser Ile Ala Phe Arg Cys	165	170	175
Ile Asn Glu Asn Leu Thr Arg Tyr Ile Ser Asn Met Asp Ile Phe Glu	180	185	190
Lys Val Asp Ala Ile Phe Asp Lys His Glu Val Gln Glu Ile Lys Glu	195	200	205
Lys Ile Leu Asn Ser Asp Tyr Asp Val Glu Asp Phe Phe Glu Gly Glu	210	215	220
Phe Phe Asn Phe Val Leu Thr Gln Glu Gly Ile Asp Val Tyr Asn Ala	225	230	235
Ile Ile Gly Gly Phe Val Thr Glu Ser Gly Glu Lys Ile Lys Gly Leu	245	250	255
Asn Glu Tyr Ile Asn Leu Tyr Asn Ala Lys Thr Lys Gln Ala Leu Pro	260	265	270
Lys Phe Lys Pro Leu Tyr Lys Gln Val Leu Ser Asp Arg Glu Ser Leu	275	280	285
Ser Phe Tyr Gly Glu Gly Tyr Thr Ser Asp Glu Glu Val Leu Glu Val	290	295	300
Phe Arg Asn Thr Leu Asn Lys Asn Ser Glu Ile Phe Ser Ser Ile Lys	305	310	315
Lys Leu Glu Lys Leu Phe Lys Asn Phe Asp Glu Tyr Ser Ser Ala Gly	325	330	335
Ile Phe Val Lys Asn Gly Pro Ala Ile Ser Thr Ile Ser Lys Asp Ile	340	345	350
Phe Gly Glu Trp Asn Leu Ile Arg Asp Lys Trp Asn Ala Glu Tyr Asp	355	360	365
Asp Ile His Leu Lys Lys Lys Ala Val Val Thr Glu Lys Tyr Glu Asp	370	375	380
Asp Arg Arg Lys Ser Phe Lys Lys Ile Gly Ser Phe Ser Leu Glu Gln	385	390	395
Leu Gln Glu Tyr Ala Asp Ala Asp Leu Ser Val Val Glu Lys Leu Lys	405	410	415
Glu Ile Ile Ile Gln Lys Val Asp Glu Ile Tyr Lys Val Tyr Gly Ser	420	425	430
Ser Glu Lys Leu Phe Asp Ala Asp Phe Val Leu Glu Lys Ser Leu Lys	435	440	445
Lys Asn Asp Ala Val Val Ala Ile Met Lys Asp Leu Leu Asp Ser Val	450	455	460
Lys Ser Phe Glu Asn Tyr Ile Lys Ala Phe Phe Gly Glu Gly Lys Glu	465	470	475
Thr Asn Arg Asp Glu Ser Phe Tyr Gly Asp Phe Val Leu Ala Tyr Asp	485	490	495
Ile Leu Leu Lys Val Asp His Ile Tyr Asp Ala Ile Arg Asn Tyr Val	500	505	510
Thr Gln Lys Pro Tyr Ser Lys Asp Lys Phe Lys Leu Tyr Phe Gln Asn	515	520	525
Pro Gln Phe Met Gly Gly Trp Asp Lys Asp Lys Glu Thr Asp Tyr Arg	530	535	540
Ala Thr Ile Leu Arg Tyr Gly Ser Lys Tyr Tyr Leu Ala Ile Met Asp	545	550	555
Lys Lys Tyr Ala Lys Cys Leu Gln Lys Ile Asp Lys Asp Asp Val Asn	565	570	575

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Gly	Asn	Tyr	Glu	Lys	Ile	Asn	Tyr	Lys	Leu	Leu	Pro	Gly	Pro	Asn	Lys
			580					585					590		
Met	Leu	Pro	Lys	Val	Phe	Phe	Ser	Lys	Lys	Trp	Met	Ala	Tyr	Tyr	Asn
		595					600					605			
Pro	Ser	Glu	Asp	Ile	Gln	Lys	Ile	Tyr	Lys	Asn	Gly	Thr	Phe	Lys	Lys
	610				615						620				
Gly	Asp	Met	Phe	Asn	Leu	Asn	Asp	Cys	His	Lys	Leu	Ile	Asp	Phe	Phe
625				630						635					640
Lys	Asp	Ser	Ile	Ser	Arg	Tyr	Pro	Lys	Trp	Ser	Asn	Ala	Tyr	Asp	Phe
			645					650						655	
Asn	Phe	Ser	Glu	Thr	Glu	Lys	Tyr	Lys	Asp	Ile	Ala	Gly	Phe	Tyr	Arg
		660						665					670		
Glu	Val	Glu	Glu	Gln	Gly	Tyr	Lys	Val	Ser	Phe	Glu	Ser	Ala	Ser	Lys
		675					680					685			
Lys	Glu	Val	Asp	Lys	Leu	Val	Glu	Glu	Gly	Lys	Leu	Tyr	Met	Phe	Gln
	690				695						700				
Ile	Tyr	Asn	Lys	Asp	Phe	Ser	Asp	Lys	Ser	His	Gly	Thr	Pro	Asn	Leu
705					710					715					720
His	Thr	Met	Tyr	Phe	Lys	Leu	Leu	Phe	Asp	Glu	Asn	Asn	His	Gly	Gln
			725						730					735	
Ile	Arg	Leu	Ser	Gly	Gly	Ala	Glu	Leu	Phe	Met	Arg	Arg	Ala	Ser	Leu
		740						745					750		
Lys	Lys	Glu	Glu	Leu	Val	Val	His	Pro	Ala	Asn	Ser	Pro	Ile	Ala	Asn
		755					760					765			
Lys	Asn	Pro	Asp	Asn	Pro	Lys	Lys	Thr	Thr	Thr	Leu	Ser	Tyr	Asp	Val
	770					775					780				
Tyr	Lys	Asp	Lys	Arg	Phe	Ser	Glu	Asp	Gln	Tyr	Glu	Leu	His	Ile	Pro
785				790						795					800
Ile	Ala	Ile	Asn	Lys	Cys	Pro	Lys	Asn	Ile	Phe	Lys	Ile	Asn	Thr	Glu
			805						810					815	
Val	Arg	Val	Leu	Leu	Lys	His	Asp	Asp	Asn	Pro	Tyr	Val	Ile	Gly	Ile
			820					825					830		
Asp	Arg	Gly	Glu	Arg	Asn	Leu	Leu	Tyr	Ile	Val	Val	Val	Asp	Gly	Lys
		835					840					845			
Gly	Asn	Ile	Val	Glu	Gln	Tyr	Ser	Leu	Asn	Glu	Ile	Ile	Asn	Asn	Phe
	850					855					860				
Asn	Gly	Ile	Arg	Ile	Lys	Thr	Asp	Tyr	His	Ser	Leu	Leu	Asp	Lys	Lys
865					870					875					880
Glu	Lys	Glu	Arg	Phe	Glu	Ala	Arg	Gln	Asn	Trp	Thr	Ser	Ile	Glu	Asn
			885					890						895	
Ile	Lys	Glu	Leu	Lys	Ala	Gly	Tyr	Ile	Ser	Gln	Val	Val	His	Lys	Ile
		900						905					910		
Cys	Glu	Leu	Val	Glu	Lys	Tyr	Asp	Ala	Val	Ile	Ala	Leu	Glu	Asp	Leu
	915						920					925			
Asn	Ser	Gly	Phe	Lys	Asn	Ser	Arg	Val	Lys	Val	Glu	Lys	Gln	Val	Tyr
	930					935					940				
Gln	Lys	Phe	Glu	Lys	Met	Leu	Ile	Asp	Lys	Leu	Asn	Tyr	Met	Val	Asp
945					950					955					960
Lys	Lys	Ser	Asn	Pro	Cys	Ala	Thr	Gly	Gly	Ala	Leu	Lys	Gly	Tyr	Gln
			965					970						975	
Ile	Thr	Asn	Lys	Phe	Glu	Ser	Phe	Lys	Ser	Met	Ser	Thr	Gln	Asn	Gly
		980						985					990		
Phe	Ile	Phe	Tyr	Ile	Pro	Ala	Trp	Leu	Thr	Ser	Lys	Ile	Asp	Pro	Ser

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995						1000					1005				
Thr	Gly	Phe	Val	Asn	Leu	Leu	Lys	Thr	Lys	Tyr	Thr	Ser	Ile	Ala	
1010						1015					1020				
Asp	Ser	Lys	Lys	Phe	Ile	Ser	Ser	Phe	Asp	Arg	Ile	Met	Tyr	Val	
1025						1030					1035				
Pro	Glu	Glu	Asp	Leu	Phe	Glu	Phe	Ala	Leu	Asp	Tyr	Lys	Asn	Phe	
1040						1045					1050				
Ser	Arg	Thr	Asp	Ala	Asp	Tyr	Ile	Lys	Lys	Trp	Lys	Leu	Tyr	Ser	
1055						1060					1065				
Tyr	Gly	Asn	Arg	Ile	Arg	Ile	Phe	Ala	Ala	Ala	Lys	Lys	Asn	Asn	
1070						1075					1080				
Val	Phe	Ala	Trp	Glu	Glu	Val	Cys	Leu	Thr	Ser	Ala	Tyr	Lys	Glu	
1085						1090					1095				
Leu	Phe	Asn	Lys	Tyr	Gly	Ile	Asn	Tyr	Gln	Gln	Gly	Asp	Ile	Arg	
1100						1105					1110				
Ala	Leu	Leu	Cys	Glu	Gln	Ser	Asp	Lys	Ala	Phe	Tyr	Ser	Ser	Phe	
1115						1120					1125				
Met	Ala	Leu	Met	Ser	Leu	Met	Leu	Gln	Met	Arg	Asn	Ser	Ile	Thr	
1130						1135					1140				
Gly	Arg	Thr	Asp	Val	Asp	Phe	Leu	Ile	Ser	Pro	Val	Lys	Asn	Ser	
1145						1150					1155				
Asp	Gly	Ile	Phe	Tyr	Asp	Ser	Arg	Asn	Tyr	Glu	Ala	Gln	Glu	Asn	
1160						1165					1170				
Ala	Ile	Leu	Pro	Lys	Asn	Ala	Asp	Ala	Asn	Gly	Ala	Tyr	Asn	Ile	
1175						1180					1185				
Ala	Arg	Lys	Val	Leu	Trp	Ala	Ile	Gly	Gln	Phe	Lys	Lys	Ala	Glu	
1190						1195					1200				
Asp	Glu	Lys	Leu	Asp	Lys	Val	Lys	Ile	Ala	Ile	Ser	Asn	Lys	Glu	
1205						1210					1215				
Trp	Leu	Glu	Tyr	Ala	Gln	Thr	Ser	Val	Lys						
1220						1225									

<210> SEQ ID NO 56
 <211> LENGTH: 36
 <212> TYPE: RNA
 <213> ORGANISM: Unknown
 <220> FEATURE:
 <223> OTHER INFORMATION: Lachnospiraceae bacterium

<400> SEQUENCE: 56

guuuccaaaga uuaaaauaaau ucuacuaagu guagau

36

<210> SEQ ID NO 57
 <211> LENGTH: 707
 <212> TYPE: PRT
 <213> ORGANISM: Unknown
 <220> FEATURE:
 <223> OTHER INFORMATION: bacteriophage

<400> SEQUENCE: 57

Met Ala Asp Thr Pro Thr Leu Phe Thr Gln Phe Leu Arg His His Leu
 1 5 10 15
 Pro Gly Gln Arg Phe Arg Lys Asp Ile Leu Lys Gln Ala Gly Arg Ile
 20 25 30
 Leu Ala Asn Lys Gly Glu Asp Ala Thr Ile Ala Phe Leu Arg Gly Lys
 35 40 45
 Ser Glu Glu Ser Pro Pro Asp Phe Gln Pro Pro Val Lys Cys Pro Ile

-continued

50	55	60
Ile Ala Cys Ser Arg Pro Leu Thr Glu Trp Pro Ile Tyr Gln Ala Ser		
65	70	75 80
Val Ala Ile Gln Gly Tyr Val Tyr Gly Gln Ser Leu Ala Glu Phe Glu		
	85	90 95
Ala Ser Asp Pro Gly Cys Ser Lys Asp Gly Leu Leu Gly Trp Phe Asp		
	100	105 110
Lys Thr Gly Val Cys Thr Asp Tyr Phe Ser Val Gln Gly Leu Asn Leu		
	115	120 125
Ile Phe Gln Asn Ala Arg Lys Arg Tyr Ile Gly Val Gln Thr Lys Val		
	130	135 140
Thr Asn Arg Asn Glu Lys Arg His Lys Lys Leu Lys Arg Ile Asn Ala		
145	150	155 160
Lys Arg Ile Ala Glu Gly Leu Pro Glu Leu Thr Ser Asp Glu Pro Glu		
	165	170 175
Ser Ala Leu Asp Glu Thr Gly His Leu Ile Asp Pro Pro Gly Leu Asn		
	180	185 190
Thr Asn Ile Tyr Cys Tyr Gln Gln Val Ser Pro Lys Pro Leu Ala Leu		
	195	200 205
Ser Glu Val Asn Gln Leu Pro Thr Ala Tyr Ala Gly Tyr Ser Thr Ser		
	210	215 220
Gly Asp Asp Pro Ile Gln Pro Met Val Thr Lys Asp Arg Leu Ser Ile		
225	230	235 240
Ser Lys Gly Gln Pro Gly Tyr Ile Pro Glu His Gln Arg Ala Leu Leu		
	245	250 255
Ser Gln Lys Lys His Arg Arg Met Arg Gly Tyr Gly Leu Lys Ala Arg		
	260	265 270
Ala Leu Leu Val Ile Val Arg Ile Gln Asp Asp Trp Ala Val Ile Asp		
	275	280 285
Leu Arg Ser Leu Leu Arg Asn Ala Tyr Trp Arg Arg Ile Val Gln Thr		
	290	295 300
Lys Glu Pro Ser Thr Ile Thr Lys Leu Leu Lys Leu Val Thr Gly Asp		
305	310	315 320
Pro Val Leu Asp Ala Thr Arg Met Val Ala Thr Phe Thr Tyr Lys Pro		
	325	330 335
Gly Ile Val Gln Val Arg Ser Ala Lys Cys Leu Lys Asn Lys Gln Gly		
	340	345 350
Ser Lys Leu Phe Ser Glu Arg Tyr Leu Asn Glu Thr Val Ser Val Thr		
	355	360 365
Ser Ile Asp Leu Gly Ser Asn Asn Leu Val Ala Val Ala Thr Tyr Arg		
	370	375 380
Leu Val Asn Gly Asn Thr Pro Glu Leu Leu Gln Arg Phe Thr Leu Pro		
385	390	395 400
Ser His Leu Val Lys Asp Phe Glu Arg Tyr Lys Gln Ala His Asp Thr		
	405	410 415
Leu Glu Asp Ser Ile Gln Lys Thr Ala Val Ala Ser Leu Pro Gln Gly		
	420	425 430
Gln Gln Thr Glu Ile Arg Met Trp Ser Met Tyr Gly Phe Arg Glu Ala		
	435	440 445
Gln Glu Arg Val Cys Gln Glu Leu Gly Leu Ala Asp Gly Ser Ile Pro		
	450	455 460
Trp Asn Val Met Thr Ala Thr Ser Thr Ile Leu Thr Asp Leu Phe Leu		
465	470	475 480

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Ala Arg Gly Gly Asp Pro Lys Lys Cys Met Phe Thr Ser Glu Pro Lys
485 490 495

Lys Lys Lys Asn Ser Lys Gln Val Leu Tyr Lys Ile Arg Asp Arg Ala
500 505 510

Trp Ala Lys Met Tyr Arg Thr Leu Leu Ser Lys Glu Thr Arg Glu Ala
515 520 525

Trp Asn Lys Ala Leu Trp Gly Leu Lys Arg Gly Ser Pro Asp Tyr Ala
530 535 540

Arg Leu Ser Lys Arg Lys Glu Glu Leu Ala Arg Arg Cys Val Asn Tyr
545 550 555 560

Thr Ile Ser Thr Ala Glu Lys Arg Ala Gln Cys Gly Arg Thr Ile Val
565 570 575

Ala Leu Glu Asp Leu Asn Ile Gly Phe Phe His Gly Arg Gly Lys Gln
580 585 590

Glu Pro Gly Trp Val Gly Leu Phe Thr Arg Lys Lys Glu Asn Arg Trp
595 600 605

Leu Met Gln Ala Leu His Lys Ala Phe Leu Glu Leu Ala His His Arg
610 615 620

Gly Tyr His Val Ile Glu Val Asn Pro Ala Tyr Thr Ser Gln Thr Cys
625 630 635 640

Pro Val Cys Arg His Cys Asp Pro Asp Asn Arg Asp Gln His Asn Arg
645 650 655

Glu Ala Phe His Cys Ile Gly Cys Gly Phe Arg Gly Asn Ala Asp Leu
660 665 670

Asp Val Ala Thr His Asn Ile Ala Met Val Ala Ile Thr Gly Glu Ser
675 680 685

Leu Lys Arg Ala Arg Gly Ser Val Ala Ser Lys Thr Pro Gln Pro Leu
690 695 700

Ala Ala Glu
705

<210> SEQ ID NO 58
<211> LENGTH: 36
<212> TYPE: RNA
<213> ORGANISM: Unknown
<220> FEATURE:
<223> OTHER INFORMATION: bacteriophage

<400> SEQUENCE: 58

ggagagaucu caaacgauug cucgaauagu cgagac

36

<210> SEQ ID NO 59
<211> LENGTH: 757
<212> TYPE: PRT
<213> ORGANISM: Unknown
<220> FEATURE:
<223> OTHER INFORMATION: bacteriophage

<400> SEQUENCE: 59

Met Pro Lys Pro Ala Val Glu Ser Glu Phe Ser Lys Val Leu Lys Lys
1 5 10 15

His Phe Pro Gly Glu Arg Phe Arg Ser Ser Tyr Met Lys Arg Gly Gly
20 25 30

Lys Ile Leu Ala Ala Gln Gly Glu Glu Ala Val Val Ala Tyr Leu Gln
35 40 45

Gly Lys Ser Glu Glu Glu Pro Pro Asn Phe Gln Pro Pro Ala Lys Cys
50 55 60

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His	Val	Val	Thr	Lys	Ser	Arg	Asp	Phe	Ala	Glu	Trp	Pro	Ile	Met	Lys	
65					70					75					80	
Ala	Ser	Glu	Ala	Ile	Gln	Arg	Tyr	Ile	Tyr	Ala	Leu	Ser	Thr	Thr	Glu	
				85					90					95		
Arg	Ala	Ala	Cys	Lys	Pro	Gly	Lys	Ser	Ser	Glu	Ser	His	Ala	Ala	Trp	
			100					105					110			
Phe	Ala	Ala	Thr	Gly	Val	Ser	Asn	His	Gly	Tyr	Ser	His	Val	Gln	Gly	
	115						120					125				
Leu	Asn	Leu	Ile	Phe	Asp	His	Thr	Leu	Gly	Arg	Tyr	Asp	Gly	Val	Leu	
	130					135					140					
Lys	Lys	Val	Gln	Leu	Arg	Asn	Glu	Lys	Ala	Arg	Ala	Arg	Leu	Glu	Ser	
145					150					155					160	
Ile	Asn	Ala	Ser	Arg	Ala	Asp	Glu	Gly	Leu	Pro	Glu	Ile	Lys	Ala	Glu	
				165					170					175		
Glu	Glu	Glu	Val	Ala	Thr	Asn	Glu	Thr	Gly	His	Leu	Leu	Gln	Pro	Pro	
			180					185					190			
Gly	Ile	Asn	Pro	Ser	Phe	Tyr	Val	Tyr	Gln	Thr	Ile	Ser	Pro	Gln	Ala	
	195						200					205				
Tyr	Arg	Pro	Arg	Asp	Glu	Ile	Val	Leu	Pro	Pro	Glu	Tyr	Ala	Gly	Tyr	
	210					215					220					
Val	Arg	Asp	Pro	Asn	Ala	Pro	Ile	Pro	Leu	Gly	Val	Val	Arg	Asn	Arg	
225					230					235				240		
Cys	Asp	Ile	Gln	Lys	Gly	Cys	Pro	Gly	Tyr	Ile	Pro	Glu	Trp	Gln	Arg	
				245					250					255		
Glu	Ala	Gly	Thr	Ala	Ile	Ser	Pro	Lys	Thr	Gly	Lys	Ala	Val	Thr	Val	
			260						265				270			
Pro	Gly	Leu	Ser	Pro	Lys	Lys	Asn	Lys	Arg	Met	Arg	Arg	Tyr	Trp	Arg	
	275						280					285				
Ser	Glu	Lys	Glu	Lys	Ala	Gln	Asp	Ala	Leu	Leu	Val	Thr	Val	Arg	Ile	
	290					295					300					
Gly	Thr	Asp	Trp	Val	Val	Ile	Asp	Val	Arg	Gly	Leu	Leu	Arg	Asn	Ala	
305					310					315				320		
Arg	Trp	Arg	Thr	Ile	Ala	Pro	Lys	Asp	Ile	Ser	Leu	Asn	Ala	Leu	Leu	
				325					330					335		
Asp	Leu	Phe	Thr	Gly	Asp	Pro	Val	Ile	Asp	Val	Arg	Arg	Asn	Ile	Val	
			340					345					350			
Thr	Phe	Thr	Tyr	Thr	Leu	Asp	Ala	Cys	Gly	Thr	Tyr	Ala	Arg	Lys	Trp	
	355						360					365				
Thr	Leu	Lys	Gly	Lys	Gln	Thr	Lys	Ala	Thr	Leu	Asp	Lys	Leu	Thr	Ala	
	370					375					380					
Thr	Gln	Thr	Val	Ala	Leu	Val	Ala	Ile	Asp	Leu	Gly	Gln	Thr	Asn	Pro	
385					390					395				400		
Ile	Ser	Ala	Gly	Ile	Ser	Arg	Val	Thr	Gln	Glu	Asn	Gly	Ala	Leu	Gln	
				405					410					415		
Cys	Glu	Pro	Leu	Asp	Arg	Phe	Thr	Leu	Pro	Asp	Asp	Leu	Leu	Lys	Asp	
			420						425				430			
Ile	Ser	Ala	Tyr	Arg	Ile	Ala	Trp	Asp	Arg	Asn	Glu	Glu	Glu	Leu	Arg	
	435						440					445				
Ala	Arg	Ser	Val	Glu	Ala	Leu	Pro	Glu	Ala	Gln	Gln	Ala	Glu	Val	Arg	
	450					455					460					
Ala	Leu	Asp	Gly	Val	Ser	Lys	Glu	Thr	Ala	Arg	Thr	Gln	Leu	Cys	Ala	
465					470					475					480	

-continued

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Lys Lys Gln Ile Lys Asp Tyr Arg Asp Ser Leu Asp Glu Leu Glu Ile		
450	455	460
Lys Ile Arg Leu Glu Ala Ile Asn Ser Leu Glu Thr Asn Gln Gln Val		
465	470	475
Glu Ile Arg Asp Leu Asp Val Phe Ser Ala Asp Arg Ala Lys Ala Asn		
485	490	495
Thr Val Asp Met Phe Asp Ile Asp Pro Asn Leu Ile Ser Trp Asp Ser		
500	505	510
Met Ser Asp Ala Arg Val Ser Thr Gln Ile Ser Asp Leu Tyr Leu Lys		
515	520	525
Asn Gly Gly Asp Glu Ser Arg Val Tyr Phe Glu Ile Asn Asn Lys Arg		
530	535	540
Ile Lys Arg Ser Asp Tyr Asn Ile Ser Gln Leu Val Arg Pro Lys Leu		
545	550	555
Ser Asp Ser Thr Arg Lys Asn Leu Asn Asp Ser Ile Trp Lys Leu Lys		
565	570	575
Arg Thr Ser Glu Glu Tyr Leu Lys Leu Ser Lys Arg Lys Leu Glu Leu		
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Ser Arg Ala Val Val Asn Tyr Thr Ile Arg Gln Ser Lys Leu Leu Ser		
595	600	605
Gly Ile Asn Asp Ile Val Ile Ile Leu Glu Asp Leu Asp Val Lys Lys		
610	615	620
Lys Phe Asn Gly Arg Gly Ile Arg Asp Ile Gly Trp Asp Asn Phe Phe		
625	630	635
Ser Ser Arg Lys Glu Asn Arg Trp Phe Ile Pro Ala Phe His Lys Ala		
645	650	655
Phe Ser Glu Leu Ser Ser Asn Arg Gly Leu Cys Val Ile Glu Val Asn		
660	665	670
Pro Ala Trp Thr Ser Ala Thr Cys Pro Asp Cys Gly Phe Cys Ser Lys		
675	680	685
Glu Asn Arg Asp Gly Ile Asn Phe Thr Cys Arg Lys Cys Gly Val Ser		
690	695	700
Tyr His Ala Asp Ile Asp Val Ala Thr Leu Asn Ile Ala Arg Val Ala		
705	710	715
Val Leu Gly Lys Pro Met Ser Gly Pro Ala Asp Arg Glu Arg Leu Gly		
725	730	735
Asp Thr Lys Lys Pro Arg Val Ala Arg Ser Arg Lys Thr Met Lys Arg		
740	745	750
Lys Asp Ile Ser Asn Ser Thr Val Glu Ala Met Val Thr Ala		
755	760	765

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 <212> TYPE: RNA
 <213> ORGANISM: Unknown
 <220> FEATURE:
 <223> OTHER INFORMATION: bacteriophage

<400> SEQUENCE: 62

accaaaacga cuaauugauug cccaguacgc ugaggac

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What is claimed is:

1. A composition comprising at least one RNA molecule comprising a cargo segment fused to a meristem transport segment (MTS), wherein the cargo segment comprises one

or more guide RNAs for an RNA-guided nuclease, wherein the RNA molecule is a substantially purified RNA molecule.

2. The composition according to claim 1, wherein the guide RNA is flanked by or comprises processing elements.

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3. The composition according to claim 2, wherein the processing elements are direct repeat sequences of a bacterial CRISPR array of the RNA-guided nuclease or are direct repeat sequences that are processed by the RNA-guided nuclease.

4. The composition according to claim 3, wherein the cargo segment comprises a plurality of guide RNAs.

5. The composition according to claim 3, wherein the guide RNAs and the direct repeat sequences of the bacterial CRISPR array are for a Cas12a or a Cas12j RNA-guided nuclease.

6. The composition according to claim 1, wherein the composition comprises both a first and a second RNA molecule each comprising a cargo segment fused to an MTS and wherein the cargo segment of the first RNA molecule comprises one or more guide RNAs for an RNA-guided nuclease.

7. The composition according to claim 6, wherein the cargo segment of the first RNA molecule comprises guide RNAs which are distinct from the guide RNAs of the second RNA molecule.

8. The composition according to claim 1, wherein the cargo segment does not contain an RNA-guided nuclease polypeptide-encoding sequence.

9. The composition according to claim 1, wherein the cargo segment further comprises an RNA-guided nuclease polypeptide-encoding sequence.

10. The composition according to claim 6, wherein the cargo segment of the first RNA molecule comprises guide RNAs and wherein the cargo segment of the second RNA molecule comprises an RNA-guided nuclease polypeptide-encoding sequence.

11. The composition according to claim 9, wherein the RNA-guided nuclease polypeptide-encoding sequence can be translated in a plant cell cytosol.

12. The composition according to claim 11, wherein the RNA molecule further comprises a polyA region.

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13. The composition according to claim 12, wherein the poly A region is 3' of the RNA-guided nuclease polypeptide-encoding sequence, and 5' of the guide RNA.

14. The composition according to claim 7, wherein the guide RNAs of the first and second RNA molecule are flanked by or comprise processing elements which are processed by different RNA-guided nucleases.

15. The composition according to claim 1, wherein the MTS comprises:

- (i) a Flowering Time (FT)-derived sequence or
- (ii) a tRNA like sequence (TLS).

16. The composition according to claim 15, wherein the MTS comprises a Flowering Time (FT)-derived sequence of SEQ ID NO: 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, or a meristem transport-competent (MTC) fragment thereof.

17. The composition according to claim 1, wherein the MTS is located 3' of the cargo segment.

18. The composition according to claim 1, further comprising an RNase inhibitor.

19. The composition according to claim 1, wherein the RNA molecule is not operably linked to a viral vector RNA and/or associated with a viral protein.

20. A method of producing a plant or plant part with an altered genome comprising:

- (i) contacting a plant or plant part with at least a first composition according to claim 1; and
- (ii) retrieving a progeny or descendant of the plant or plant part, wherein the progeny or descendant has an altered genome.

21. The method according to claim 20, wherein contacting comprises phloem loading.

22. The method according to claim 20, wherein the contacting with the composition occurs at the vegetative stage of the plant life cycle.

23. A plant or plant part comprising an altered genome made by the method of claim 20.

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